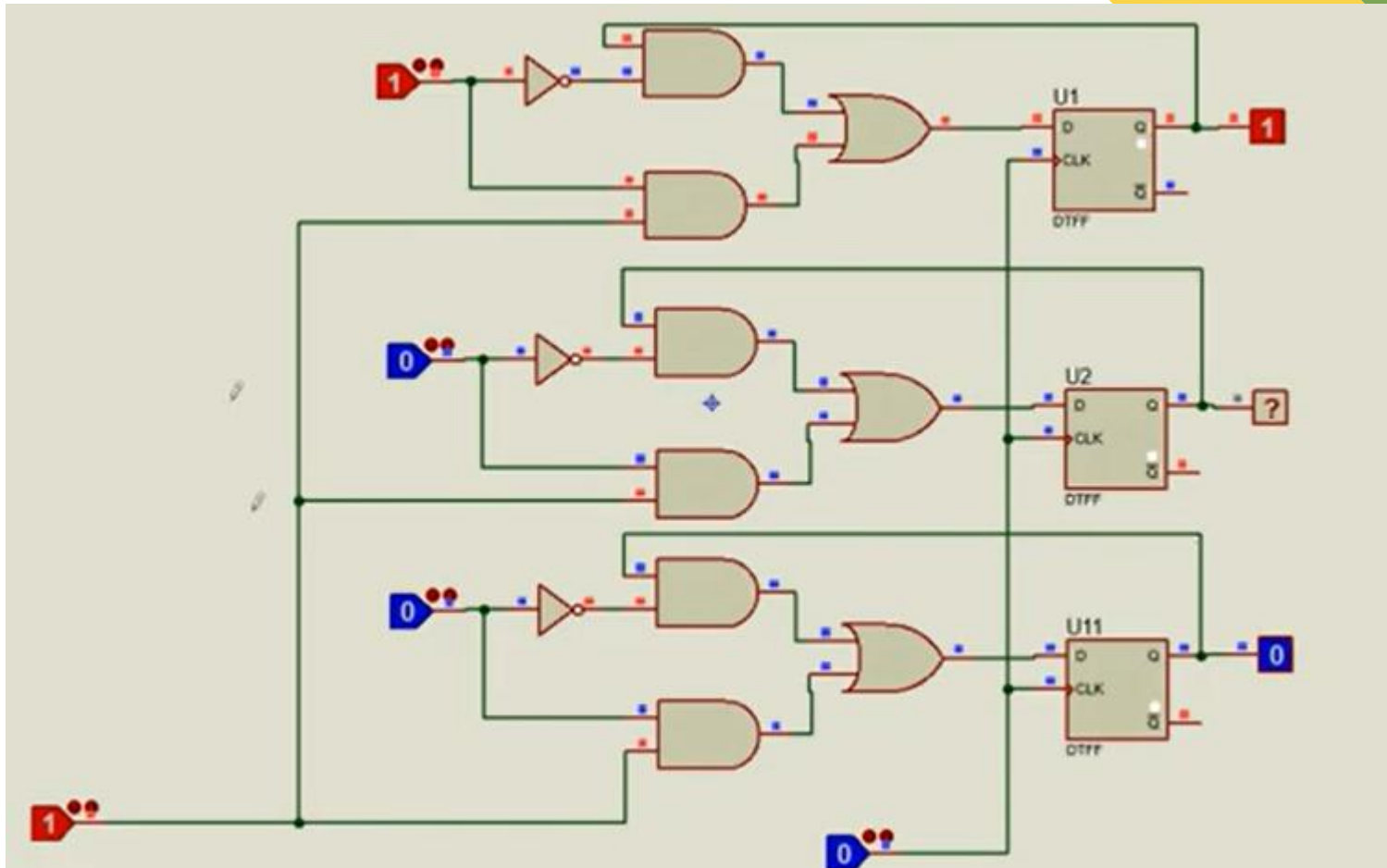




Contadores

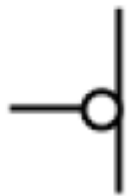
Registrador deslocamento - SIPO



Simbologia para sinais de controle



NÍVEL '1'



NÍVEL '0'



SUBIDA

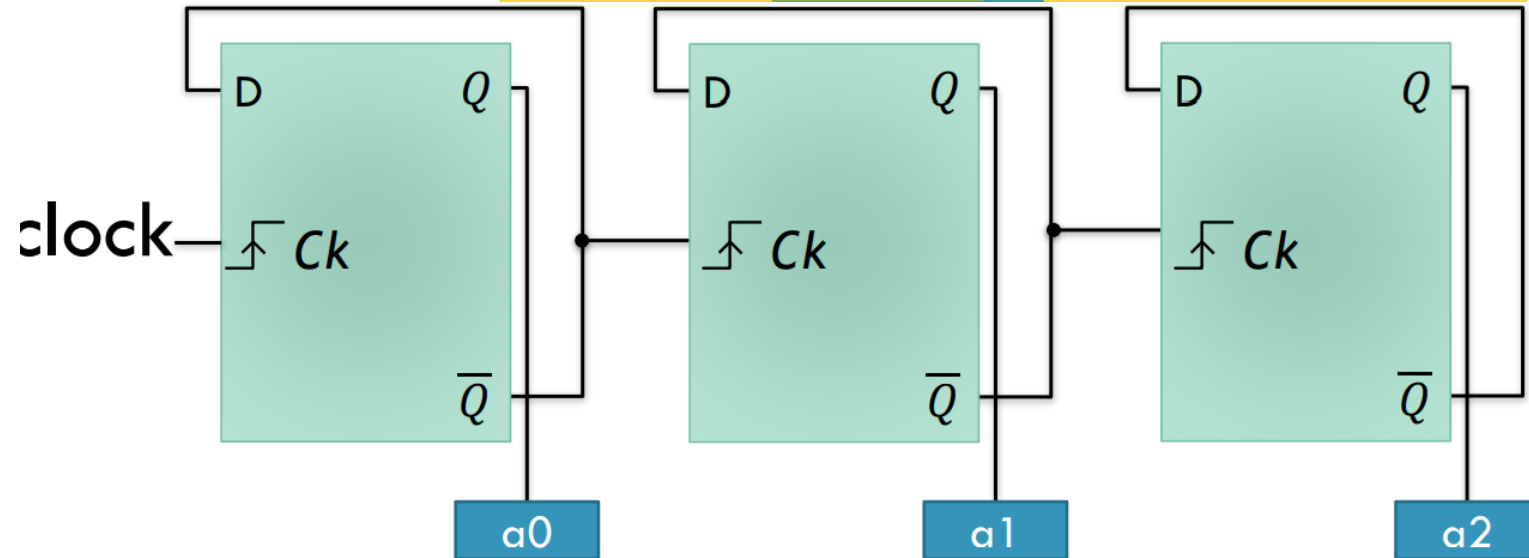


DESCIDA

SIMBOLOGIA DE NÍVEIS VÁLIDOS PARA TODOS OS SINAIS.
SIMBOLOGIA DE RAMPA APENAS PARA CONROLE

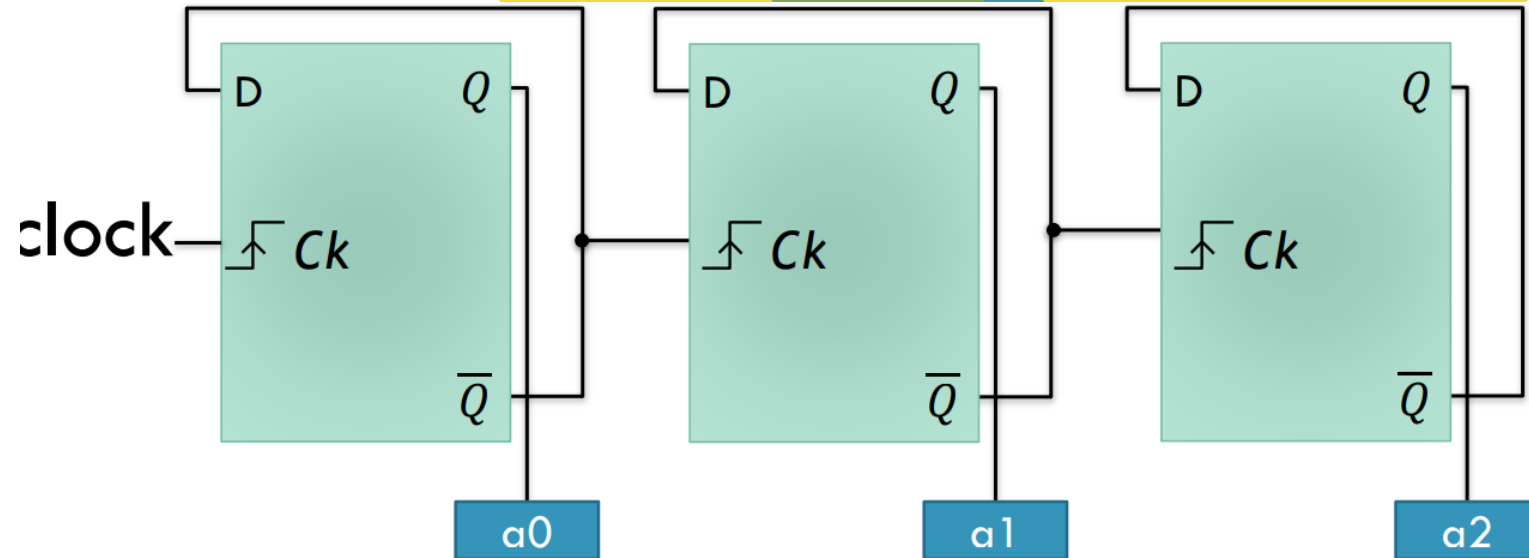
Flip Flop tipo D - Aplicação

- O que faz o circuito abaixo?
- entrada:
 - Ck ☐
- saídas:
 - a_2, a_1, a_0
- Suponha que o estado inicial de cada saída é 0.



Flip Flop tipo D - Aplicação

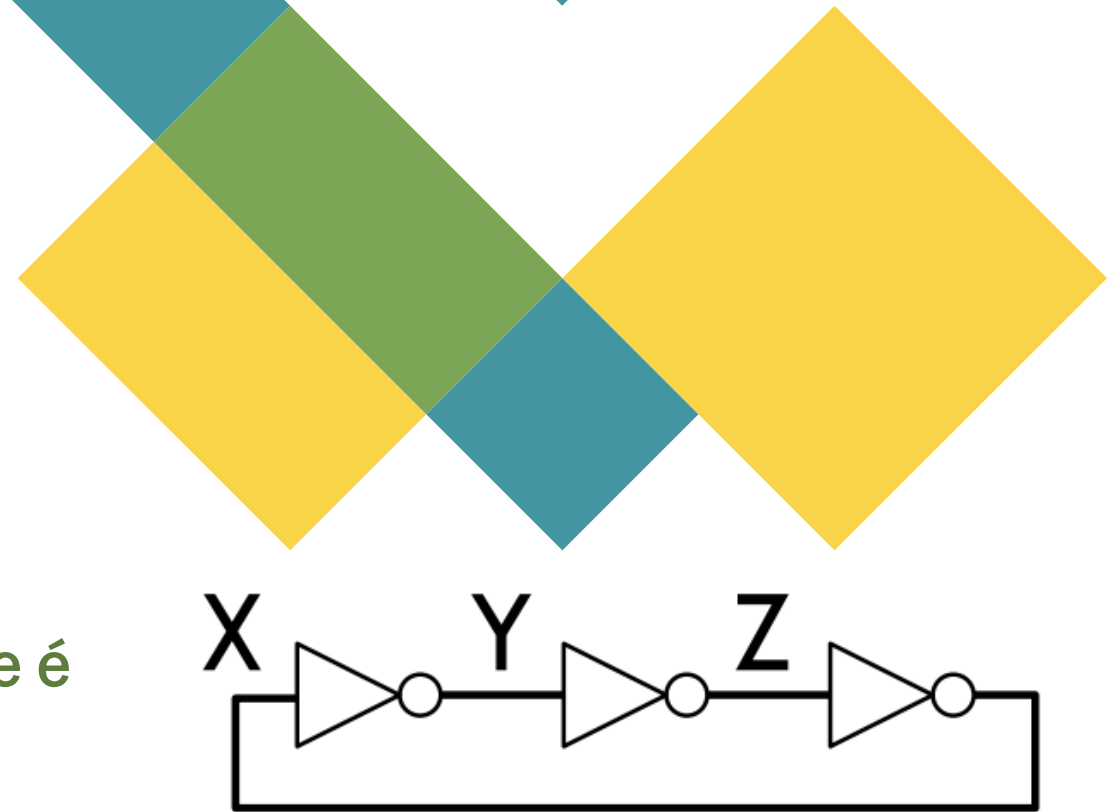
- O que faz o circuito abaixo?
- entrada:
 - Ck
- saídas:
 - a2, a1, a0
- Suponha que o estado inicial de cada saída é 0.



Contador de 3 bits

CIRCUITOS INSTÁVEIS

- Analisando esse circuito observamos:
- Supondo que $X=0$, então $Y=1$, $Z=0$ e $X=1$, o que é inconsistente com nossa suposição inicial.
- Esse circuito não possui um estado estável, por isso é chamado de instável



Contadores

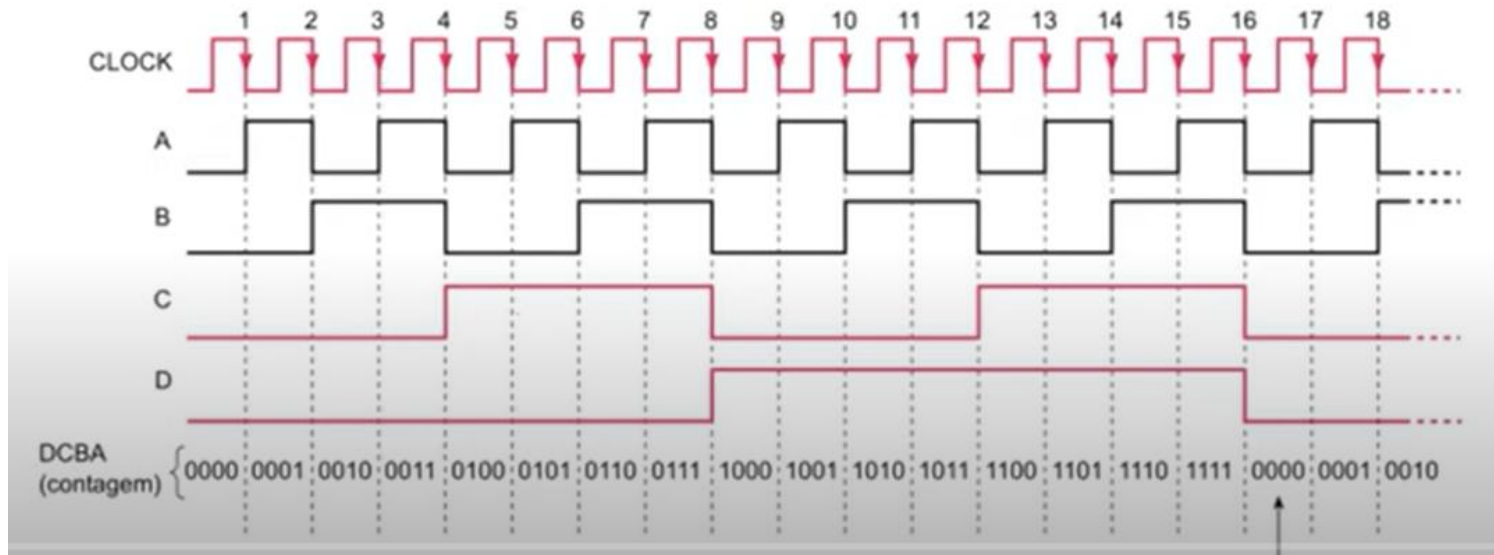
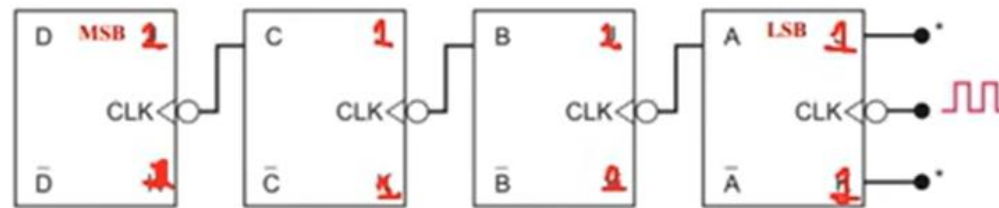
- Assíncrono
- O clock esta ligado a apenas 01 flip flop e os outros estão em cascata
- Síncrono
 - Todos os flip flop estão ligados a um clock de referência



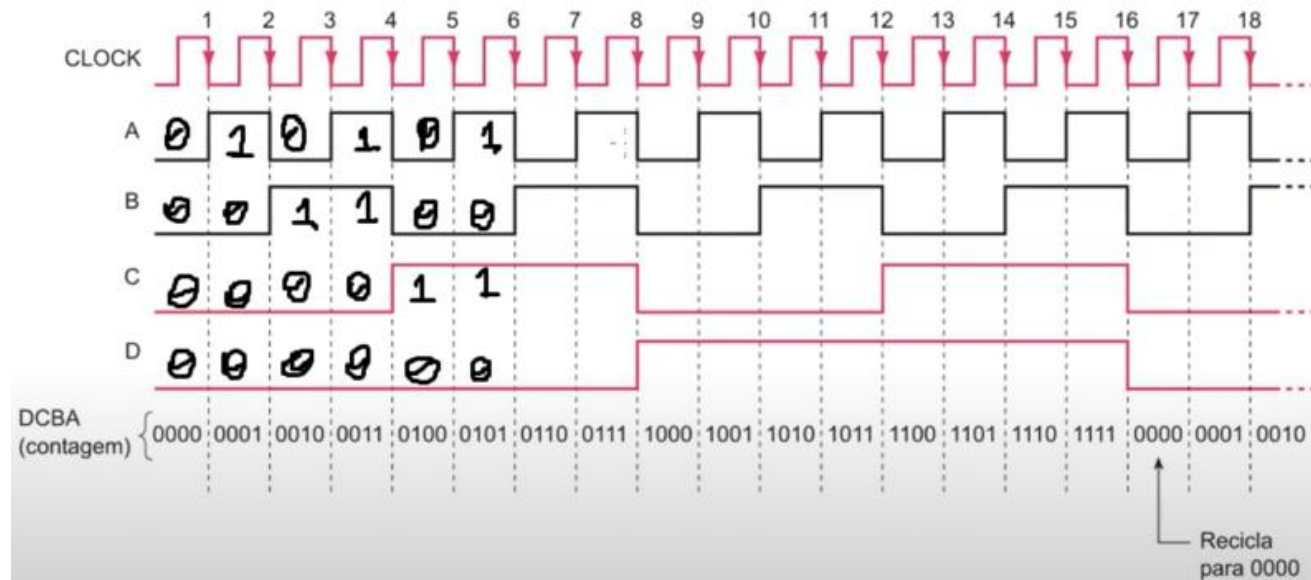
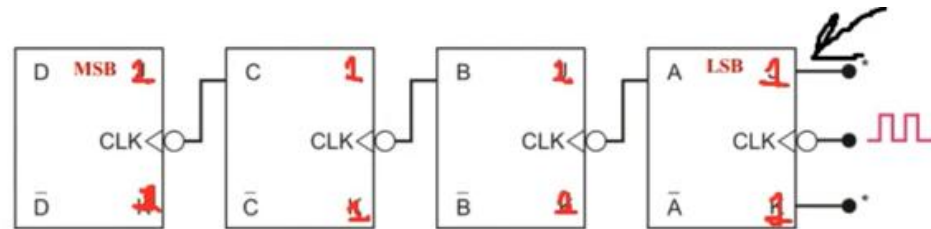


Contadores Assíncronos

Usando Flip Flop (Tipo T)

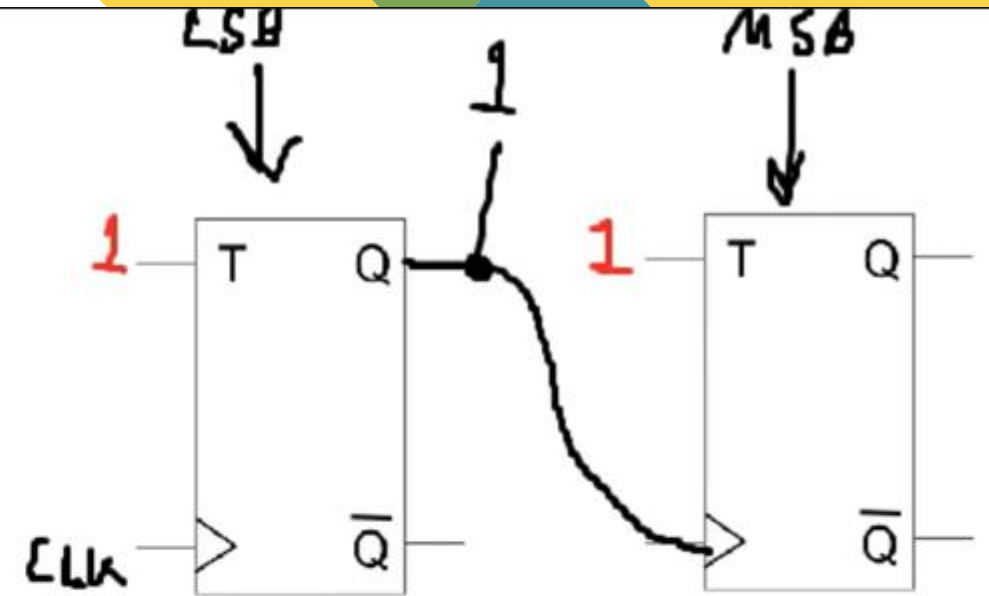


Análise

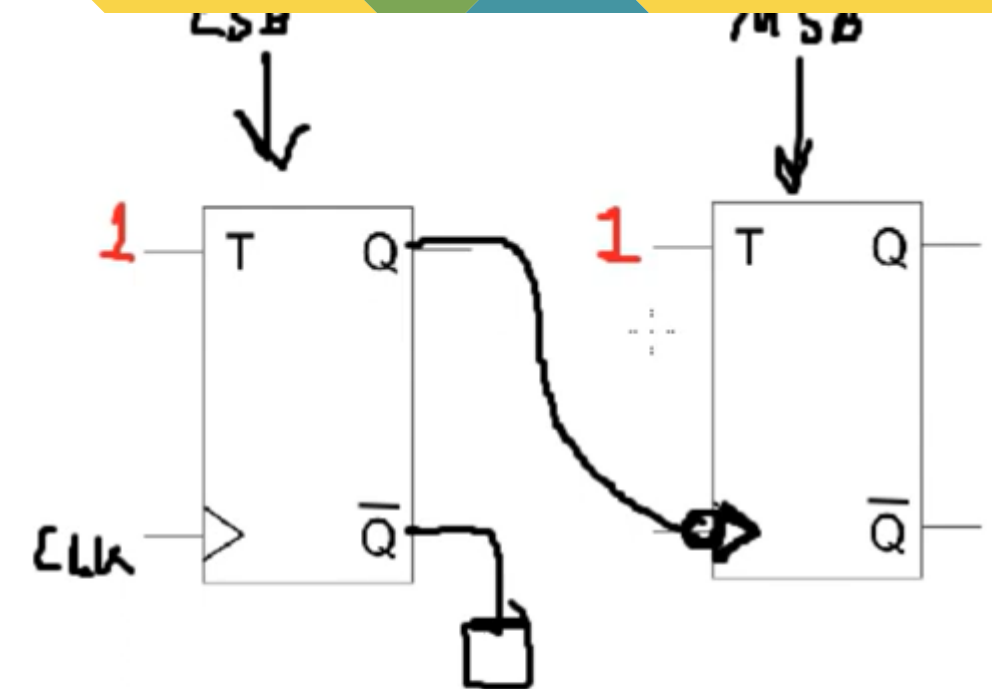


↙

tipo de contador assíncrono	saída do FF anterior	saída de contagem	clock base
CRESCENTE	Q	Q	↓
CRESCENTE	Q!	Q	↑
DESCRESCENTE	Q	Q!	↓
DESCRESCENTE	Q!	Q!	↑
DESCRESCENTE	Q!	Q	↓
DESCRESCENTE	Q	Q	↑

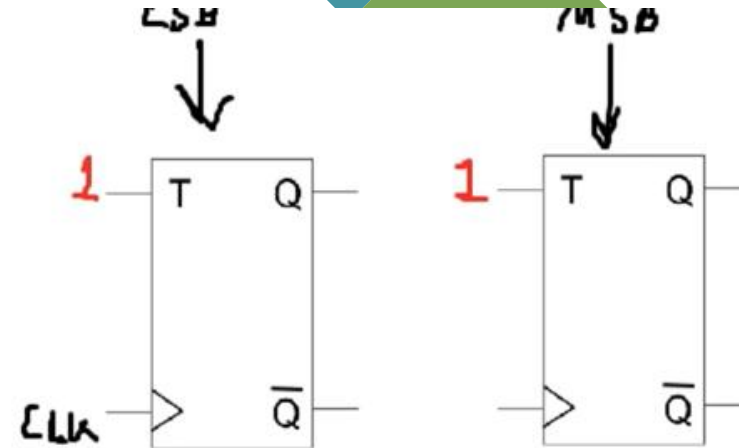


tipo de contador assíncrono	saída do FF anterior	saída de contagem	clock base
CRESCENTE	Q	Q	↓
CRESCENTE	Q!	Q	↑
DESCRESCENTE	Q	Q!	↓ ✖
DESCRESCENTE	Q!	Q!	↑ ✖
DESCRESCENTE	Q!	Q	↓
DESCRESCENTE	Q	Q	↑



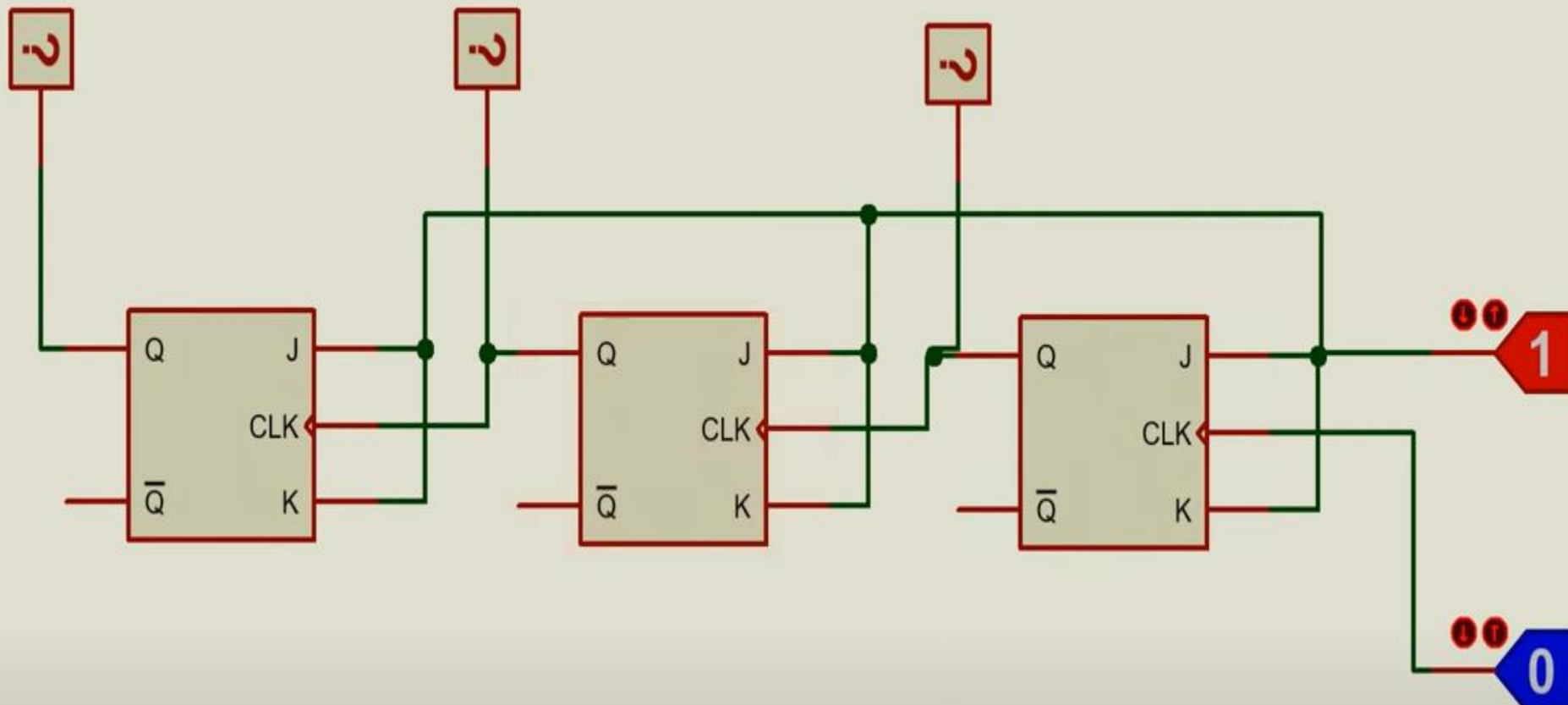
Contador assíncrono

tipo de contador assíncrono	saída do FF anterior	saída de contagem	clock base
CRESCENTE	Q	Q	↓
CRESCENTE	Q!	Q	↑
DESCRESCENTE	Q	Q!	↓ *
DESCRESCENTE	Q!	Q!	↑ *
DESCRESCENTE	Q!	Q	↓
DESCRESCENTE	Q	Q	↑

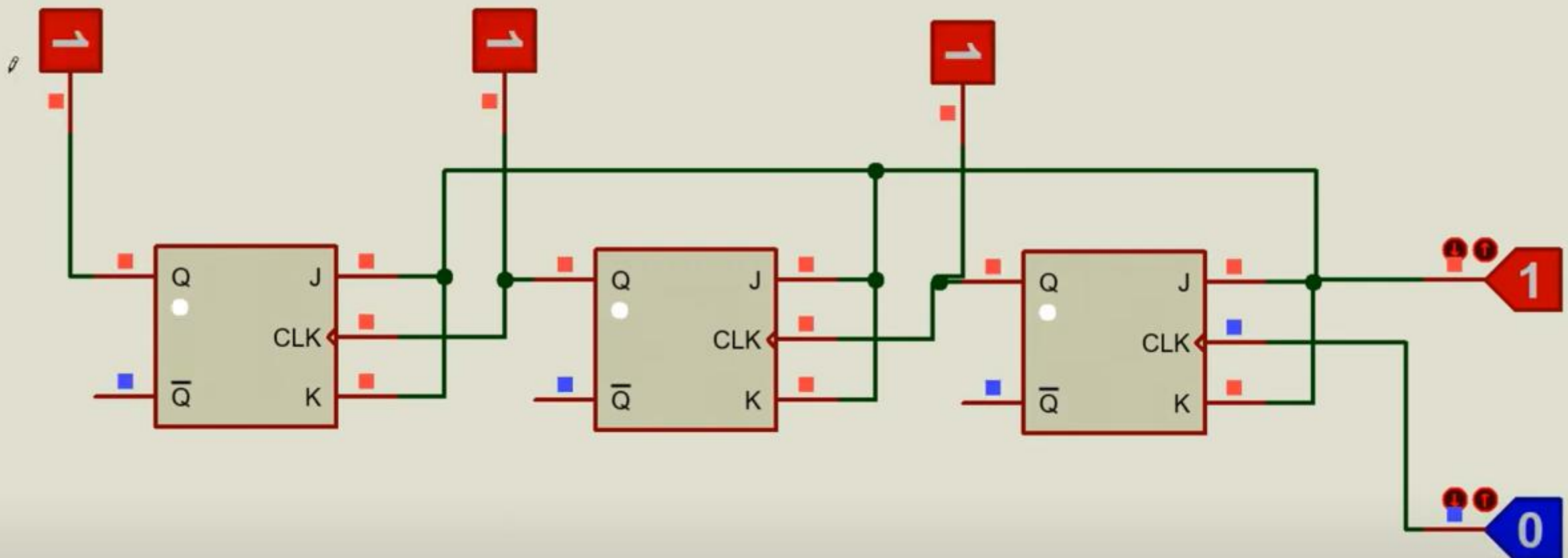


$$* \rightarrow m = 2^n - 1$$

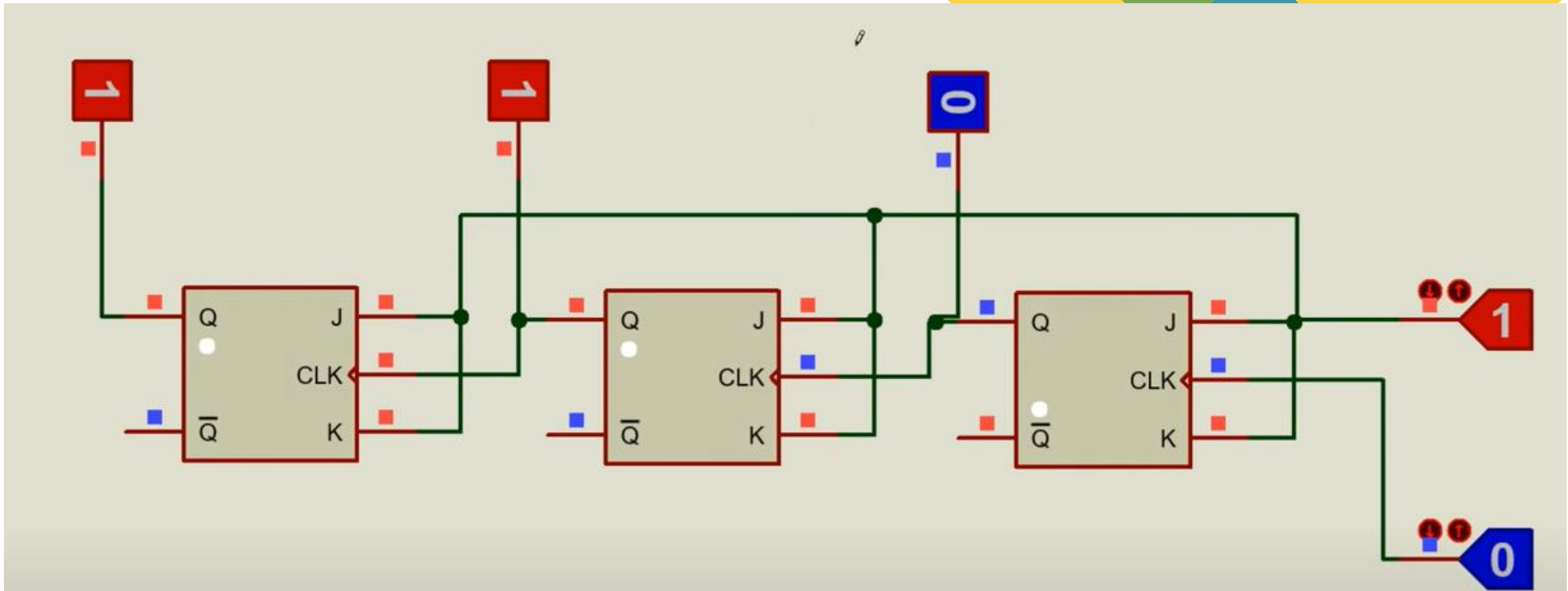
4 bits $\rightarrow$ 4 FF



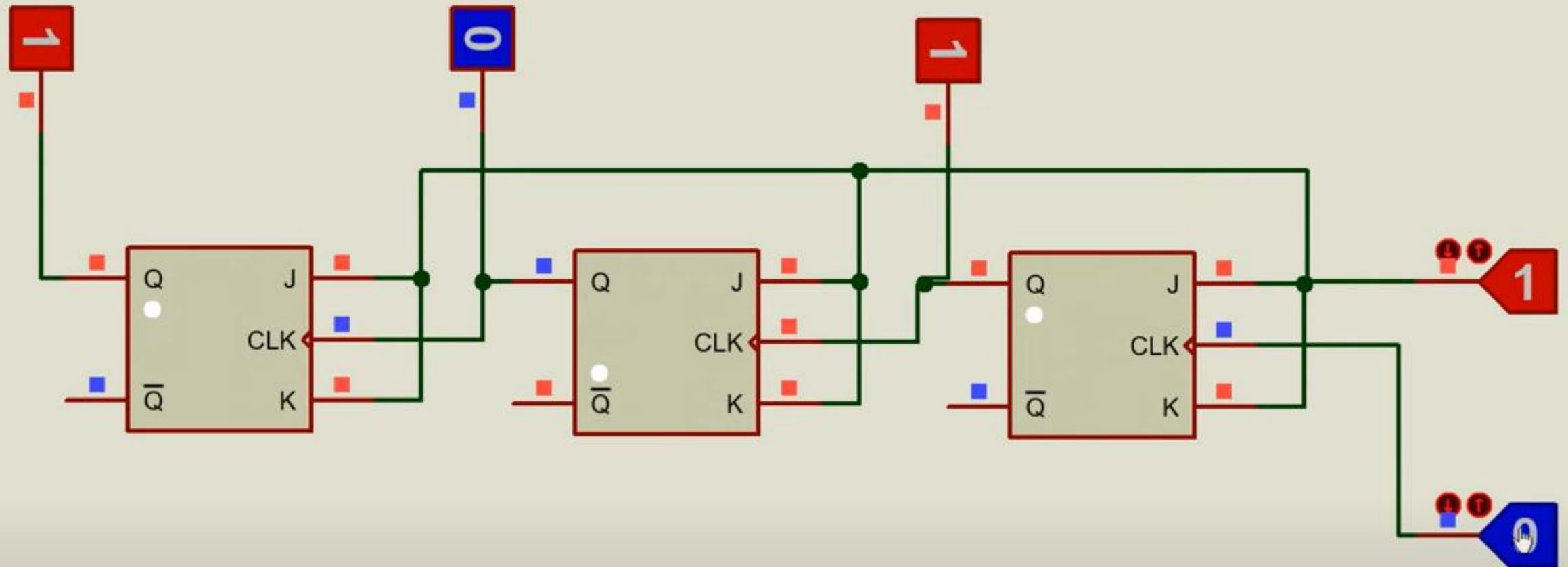
- Bora de subida



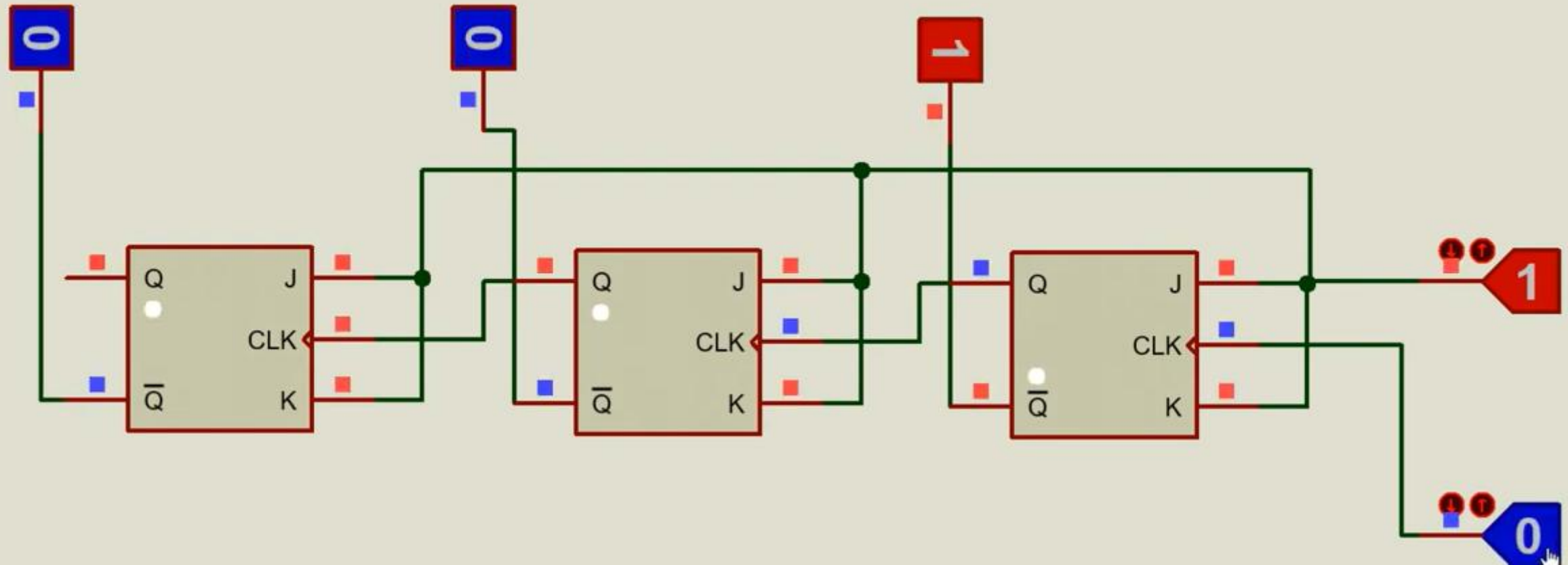
Contador decreciente

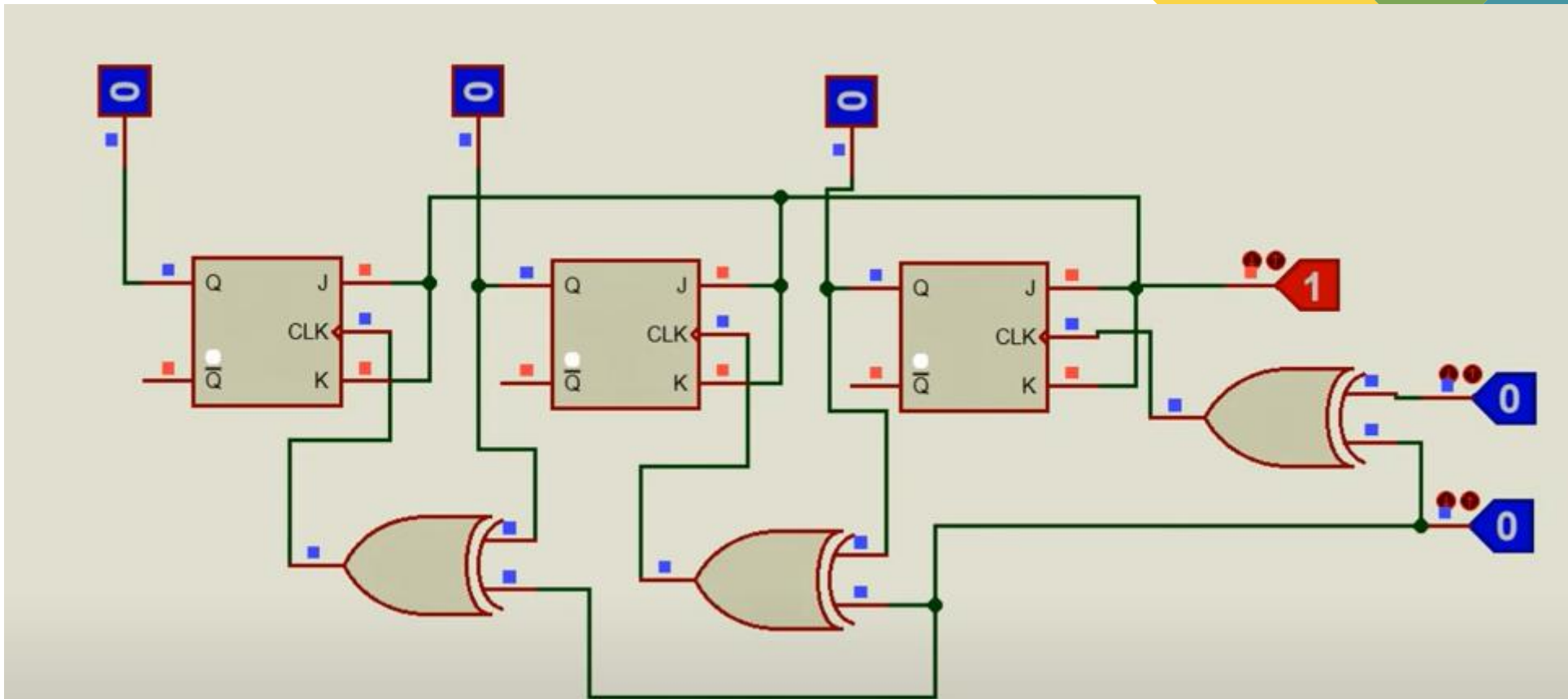
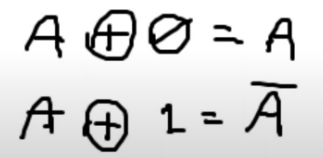


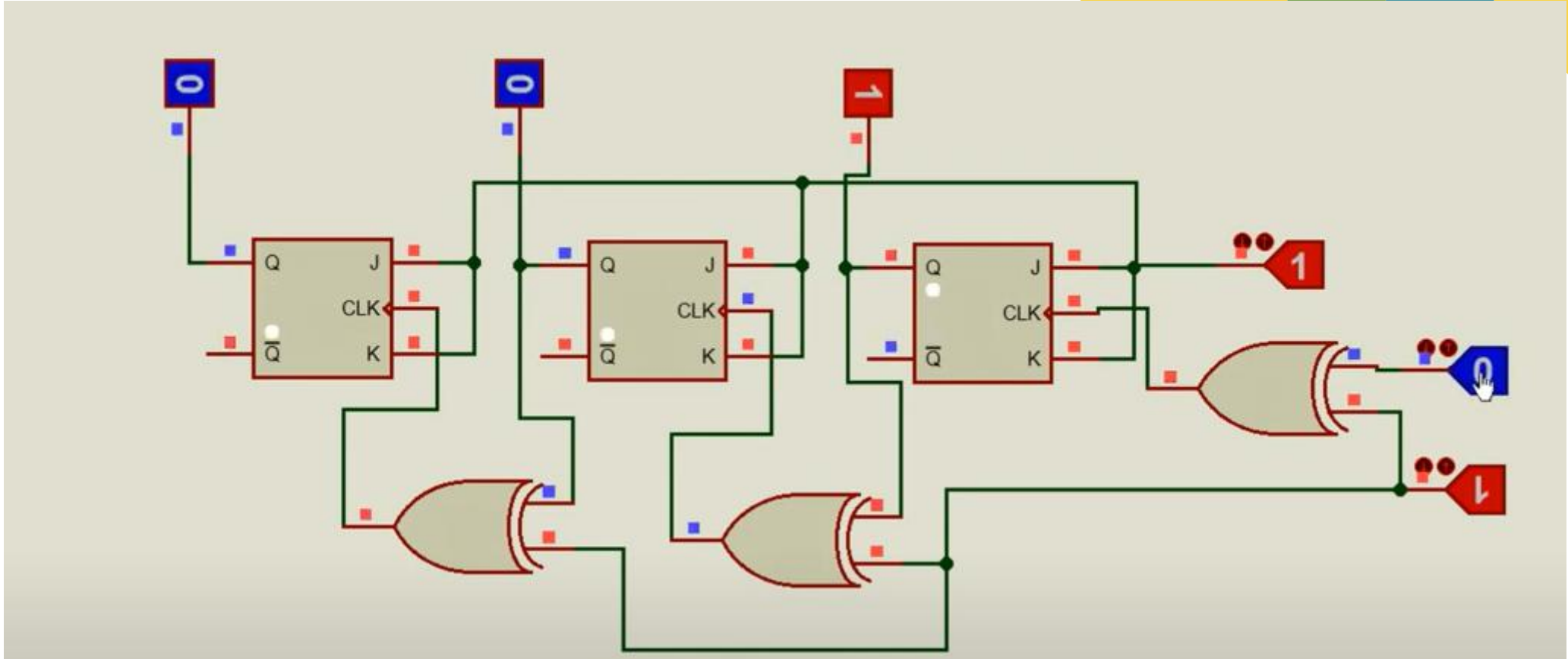
Contador Decrescente



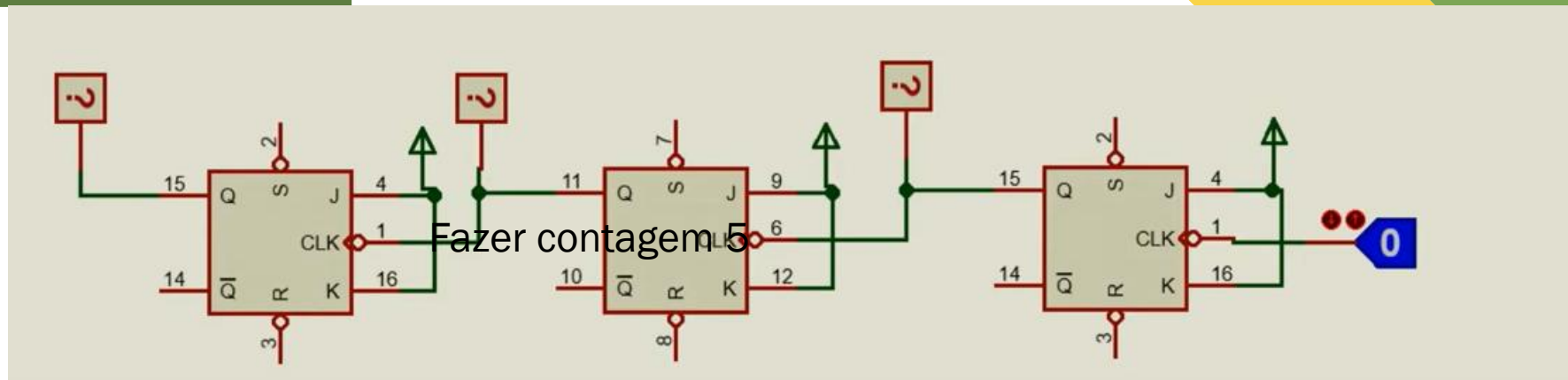
Contador crescente

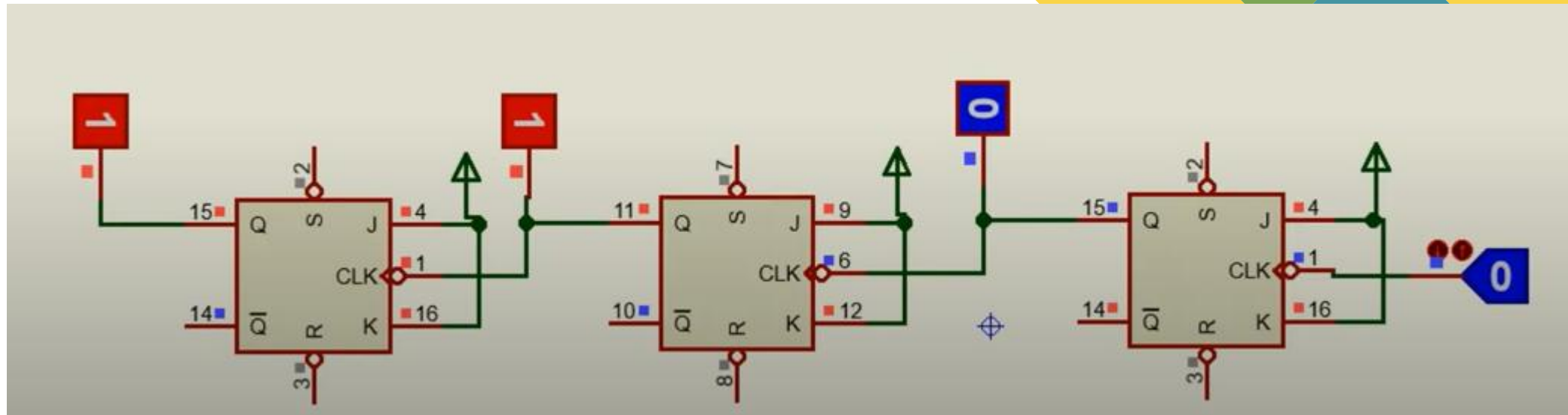




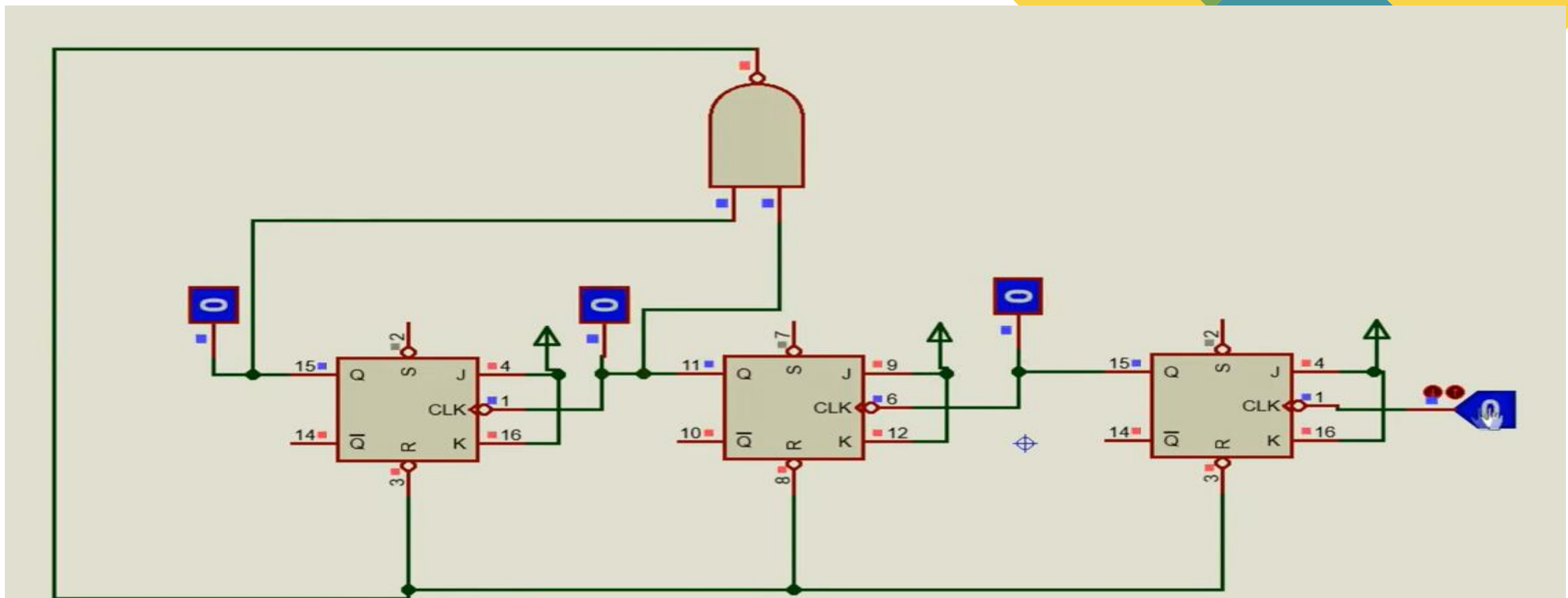


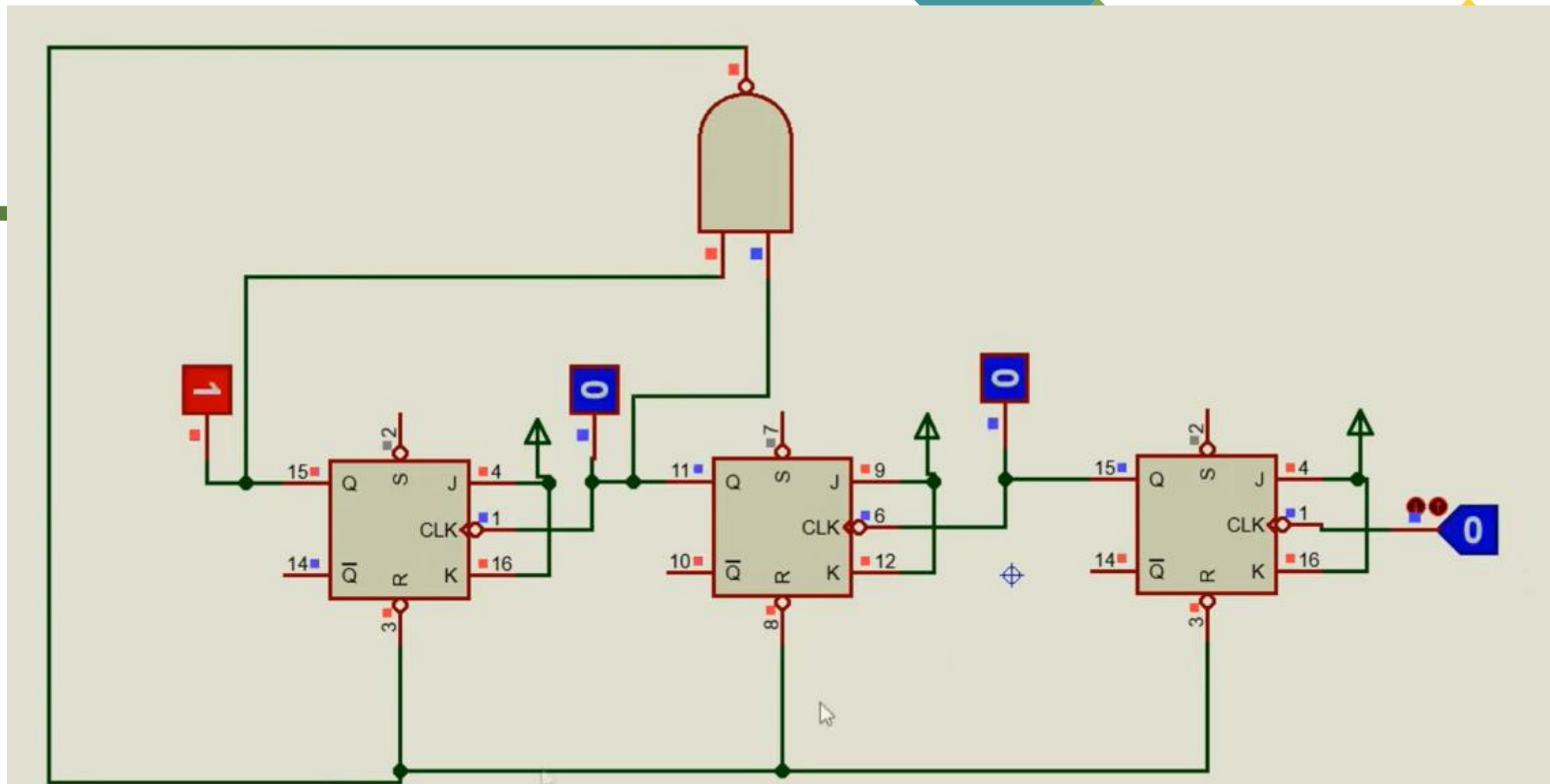
Contadores com contagem menor (igual a 5)

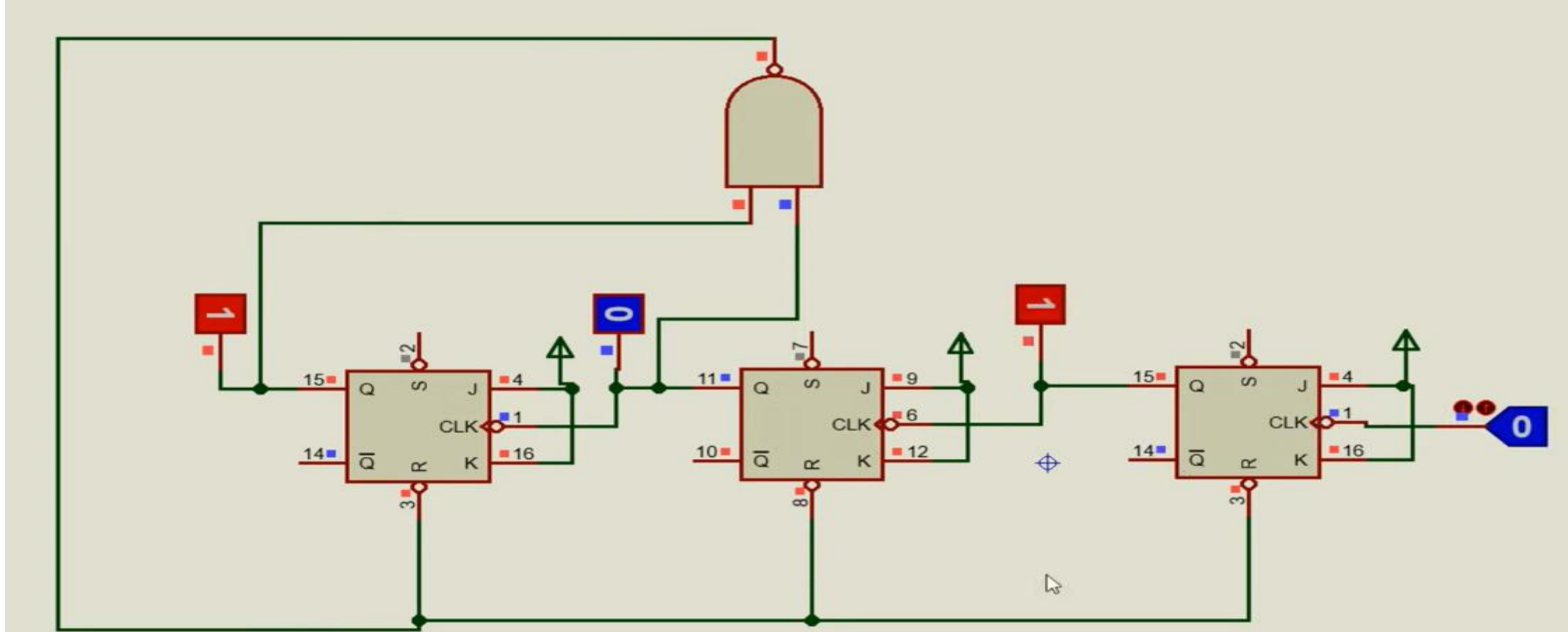




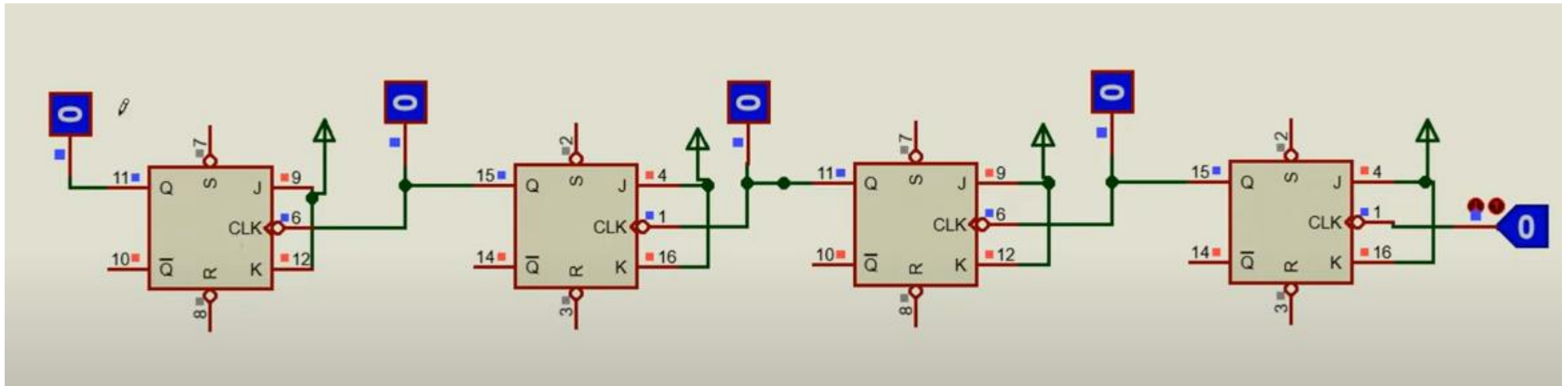
Contra até 5 → passa rapidamente por 6



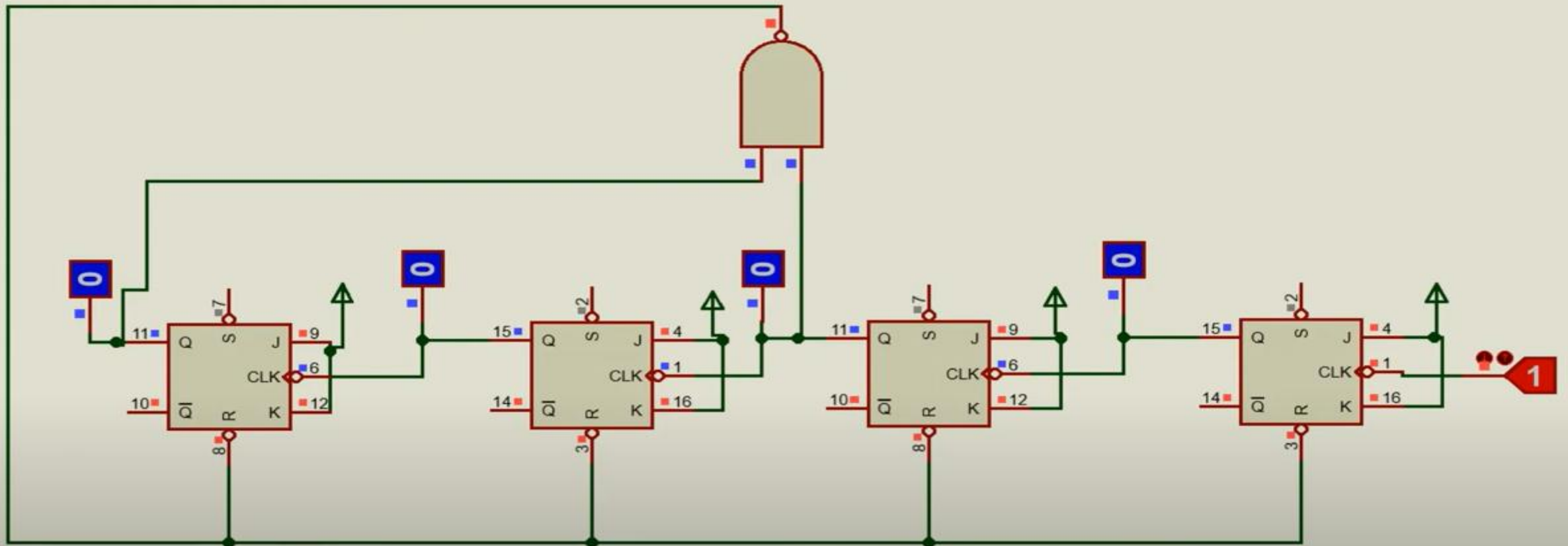




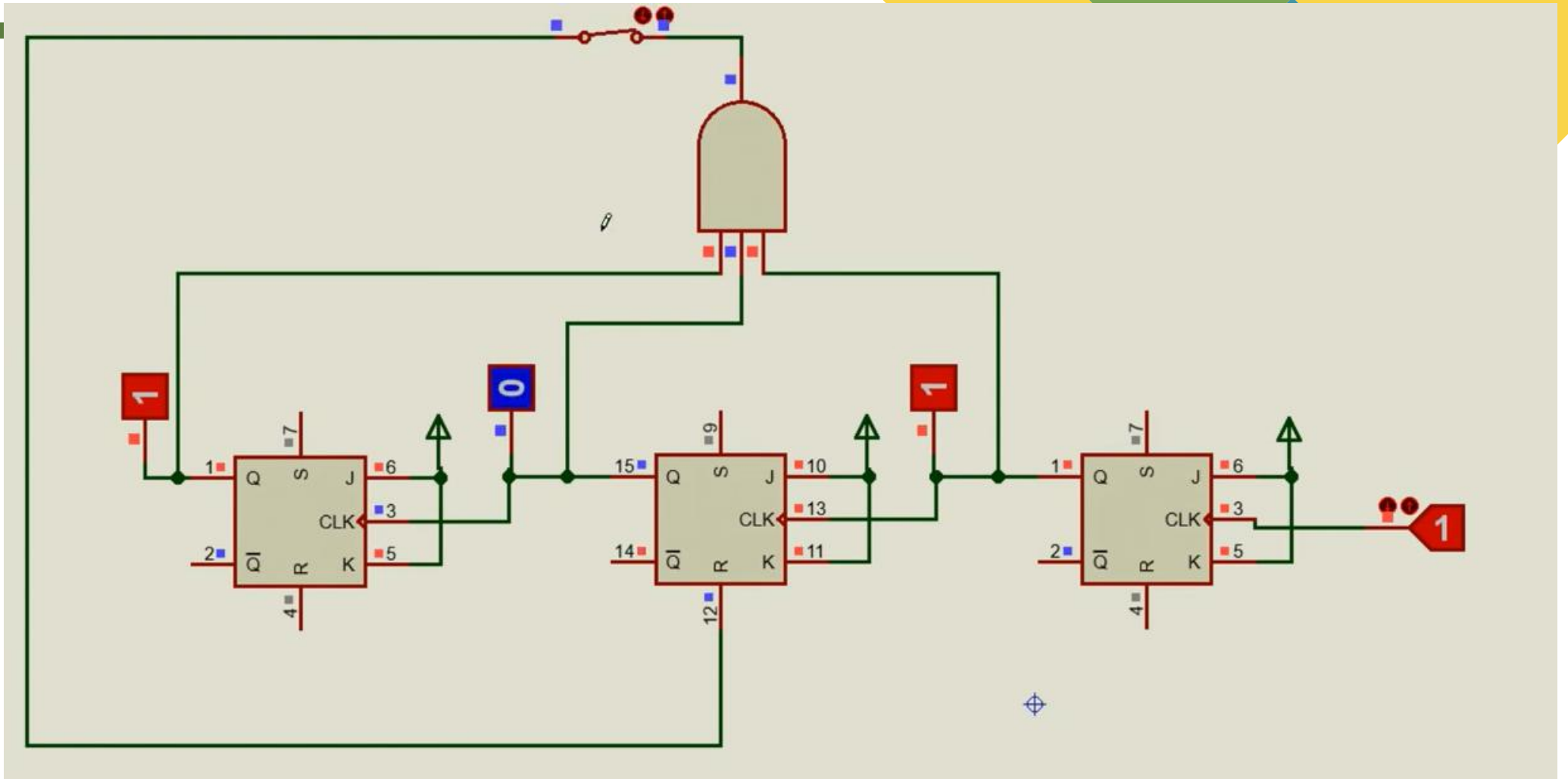
Outro exemplo



Qual o numero máximo de contagem?



Contagem decrescente (5-0)

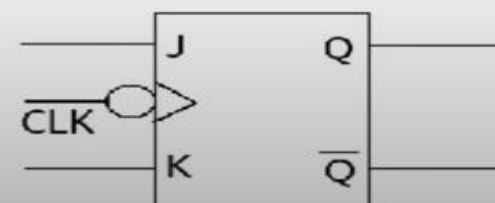
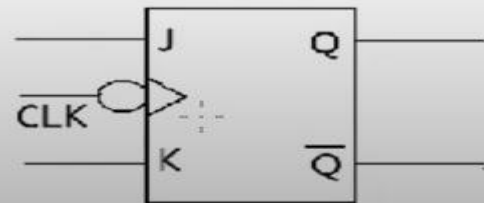




Contadores Síncronos

CONTADOR SÍNCRONO

pulsos	Q3	Q2	Q1	Q0
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10	1	0	1	0
11	1	0	1	1
12	1	1	0	0
13	1	1	0	1
14	1	1	1	0
15	1	1	1	1



CONTADOR SÍNCRONO

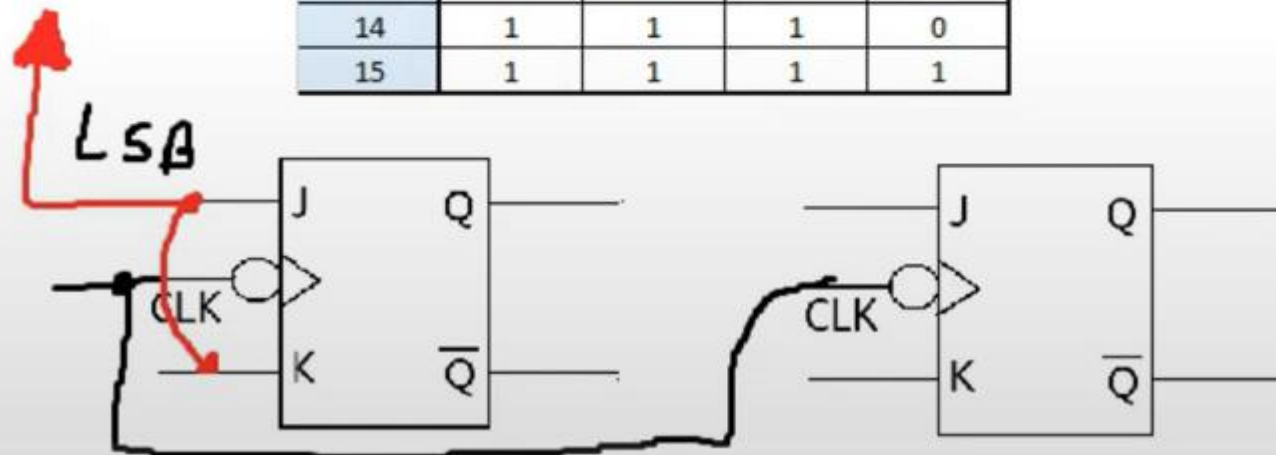
pulsos	Q3	Q2	Q1	Q0
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10	1	0	1	0
11	1	0	1	1
12	1	1	0	0
13	1	1	0	1
14	1	1	1	0
15	1	1	1	1



1ª Obs

CONTADOR SÍNCRONO

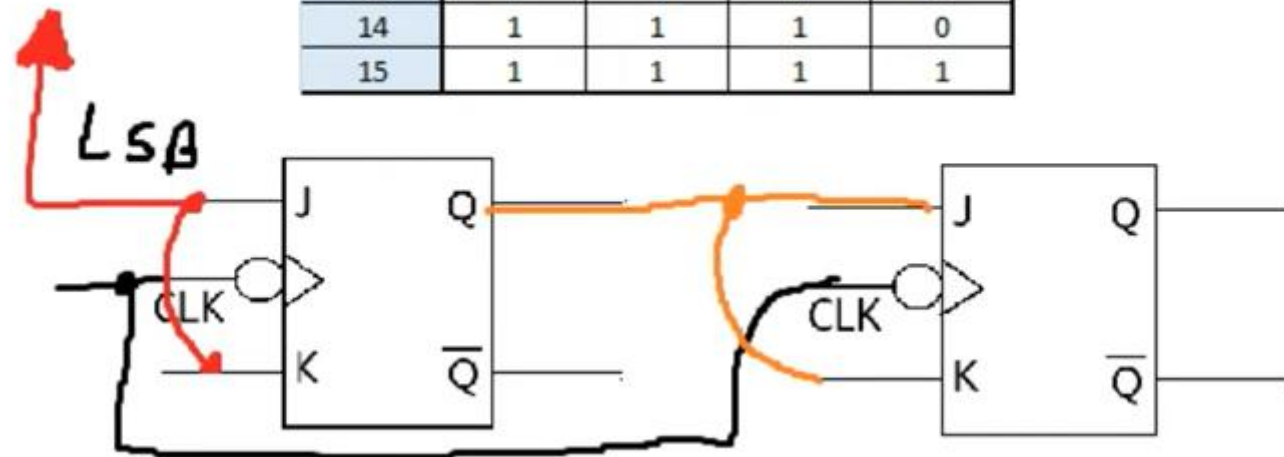
	MSB		LSB	
pulsos	Q3	Q2	Q1	Q0
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10	1	0	1	0
11	1	0	1	1
12	1	1	0	0
13	1	1	0	1
14	1	1	1	0
15	1	1	1	1



CONTADOR SÍNCRONO

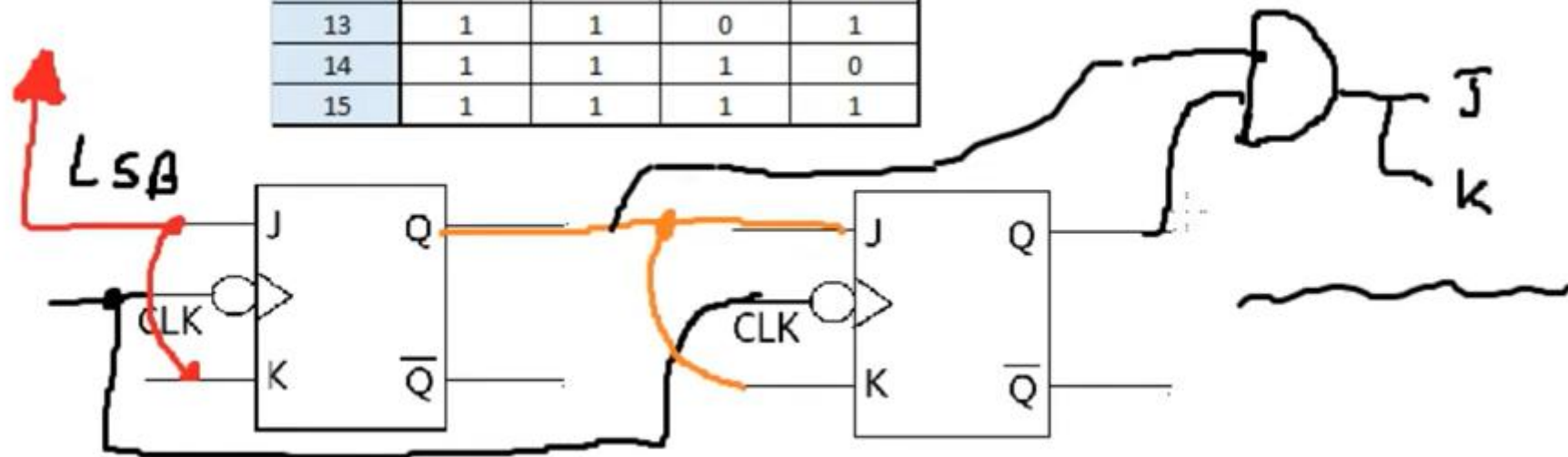
MSB *LSB*

pulsos	Q3	Q2	Q1	Q0
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10	1	0	1	0
11	1	0	1	1
12	1	1	0	0
13	1	1	0	1
14	1	1	1	0
15	1	1	1	1

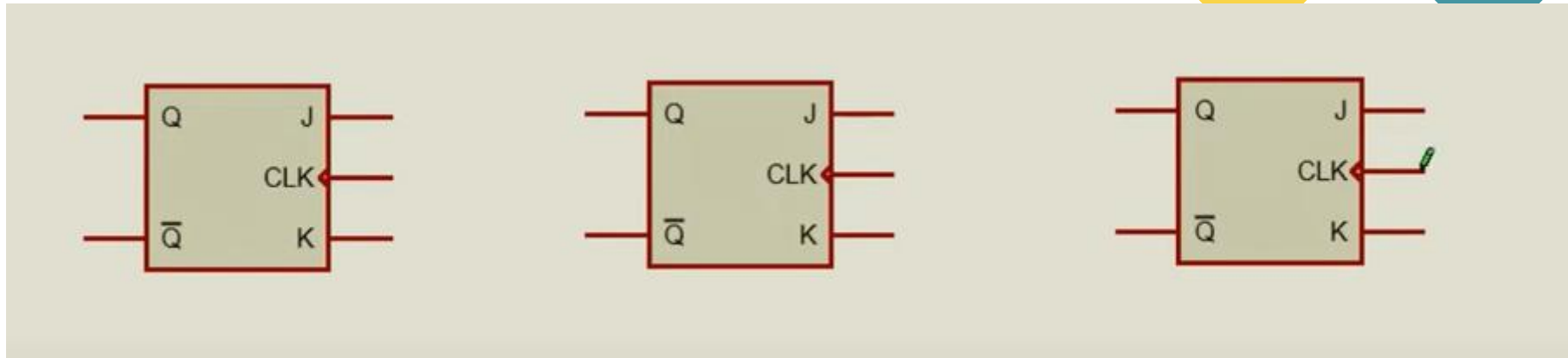


CONTADOR SÍNCRONO

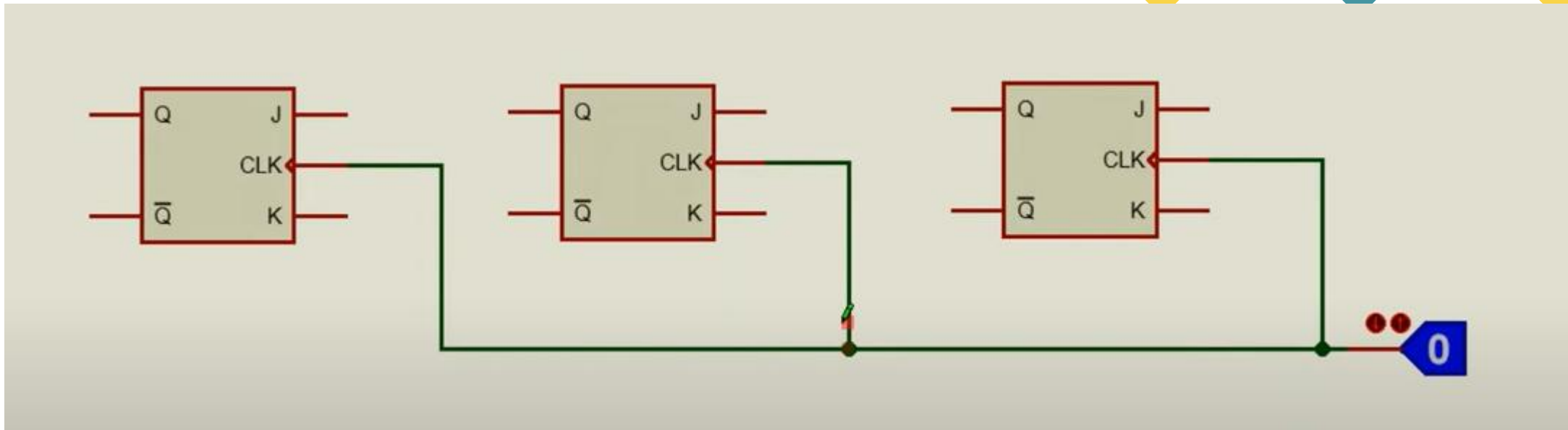
	MSB		LSB	
pulsos	Q3	Q2	Q1	Q0
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10	1	0	1	0
11	1	0	1	1
12	1	1	0	0
13	1	1	0	1
14	1	1	1	0
15	1	1	1	1



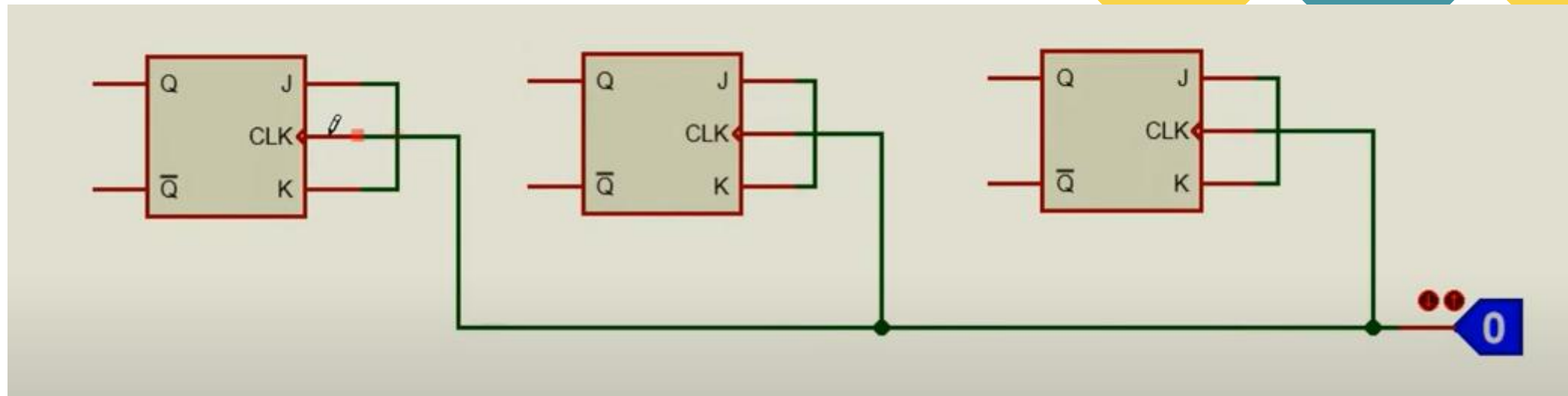
Montagem de um contador síncrono



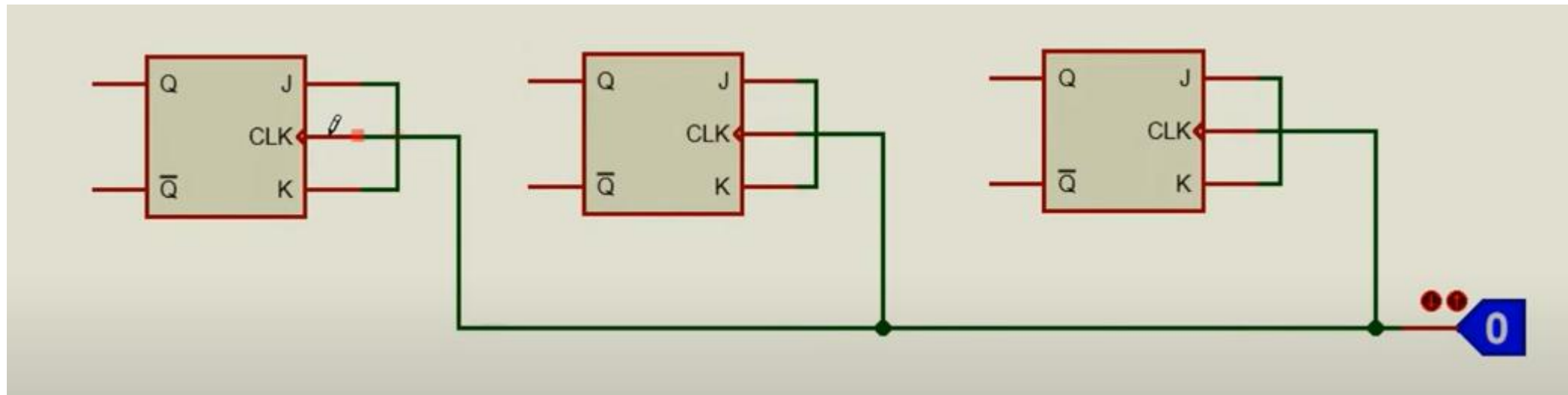
1ª) Etapa

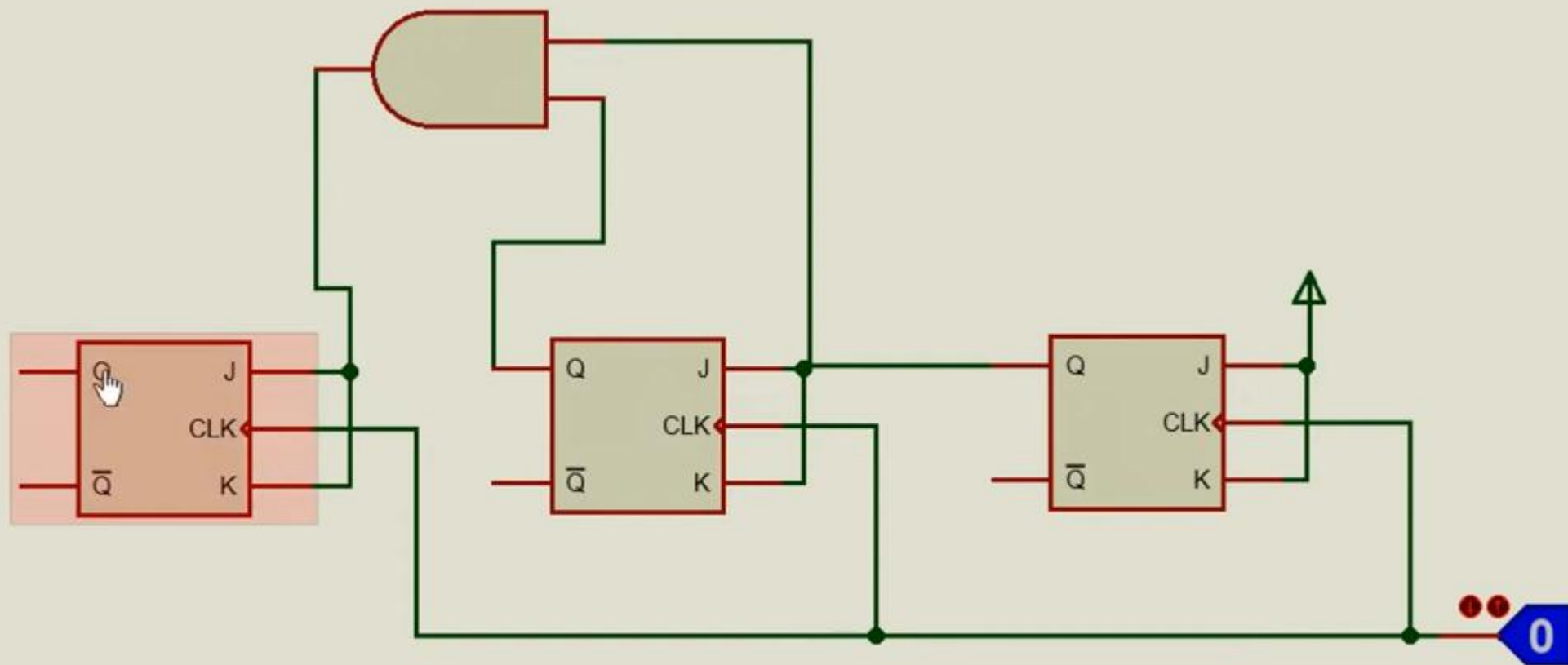


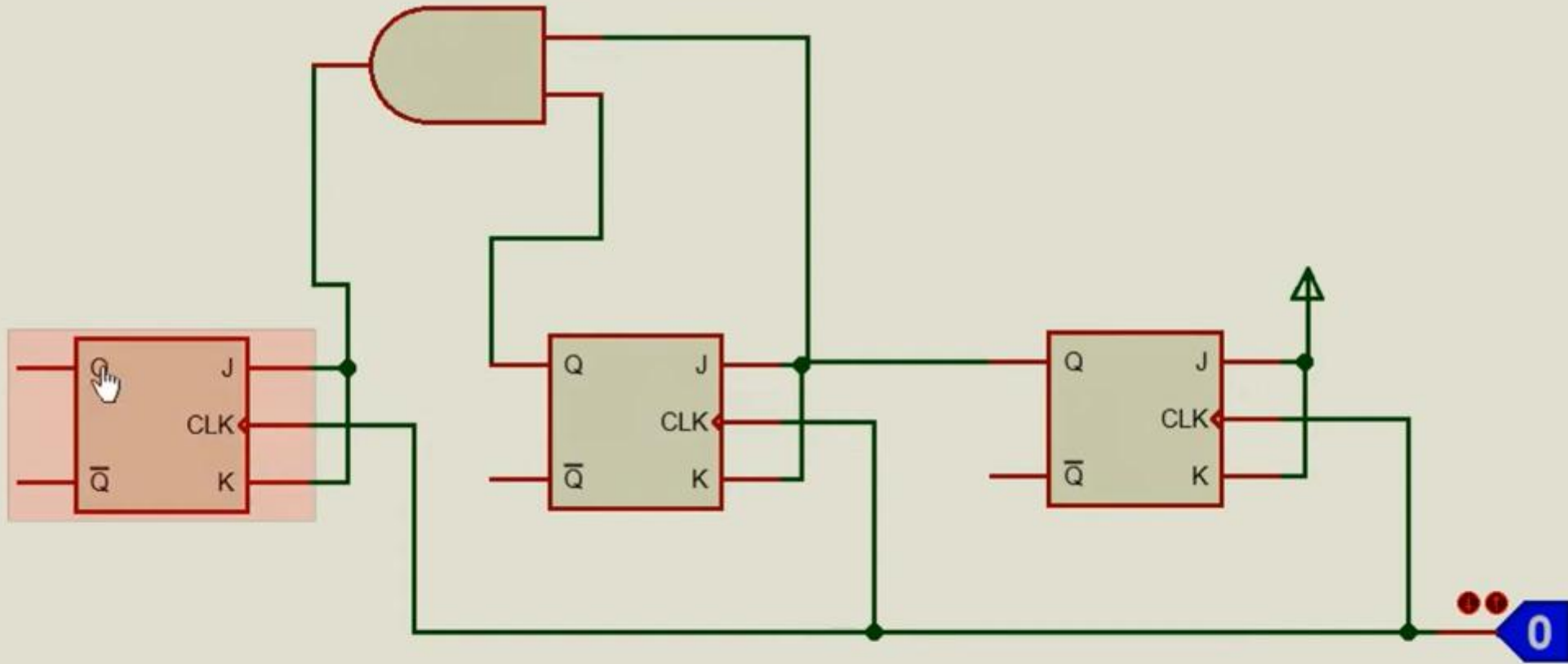
2ª Etapa

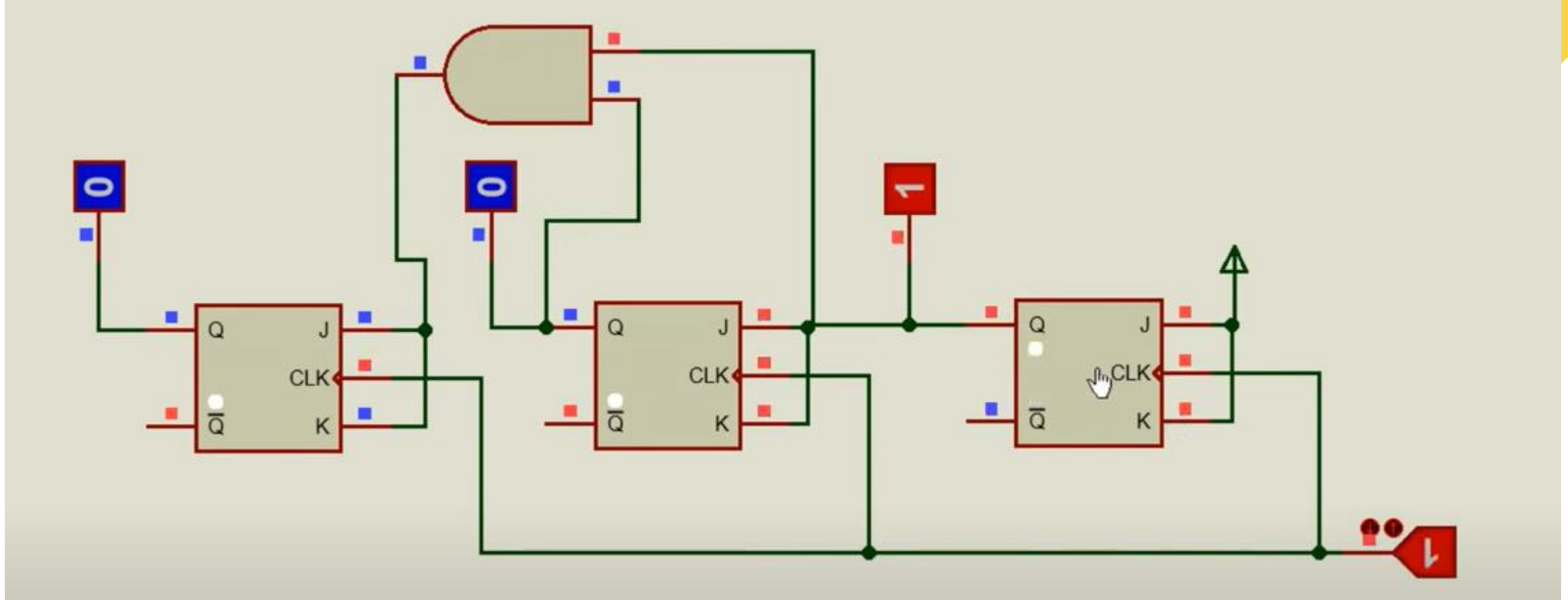


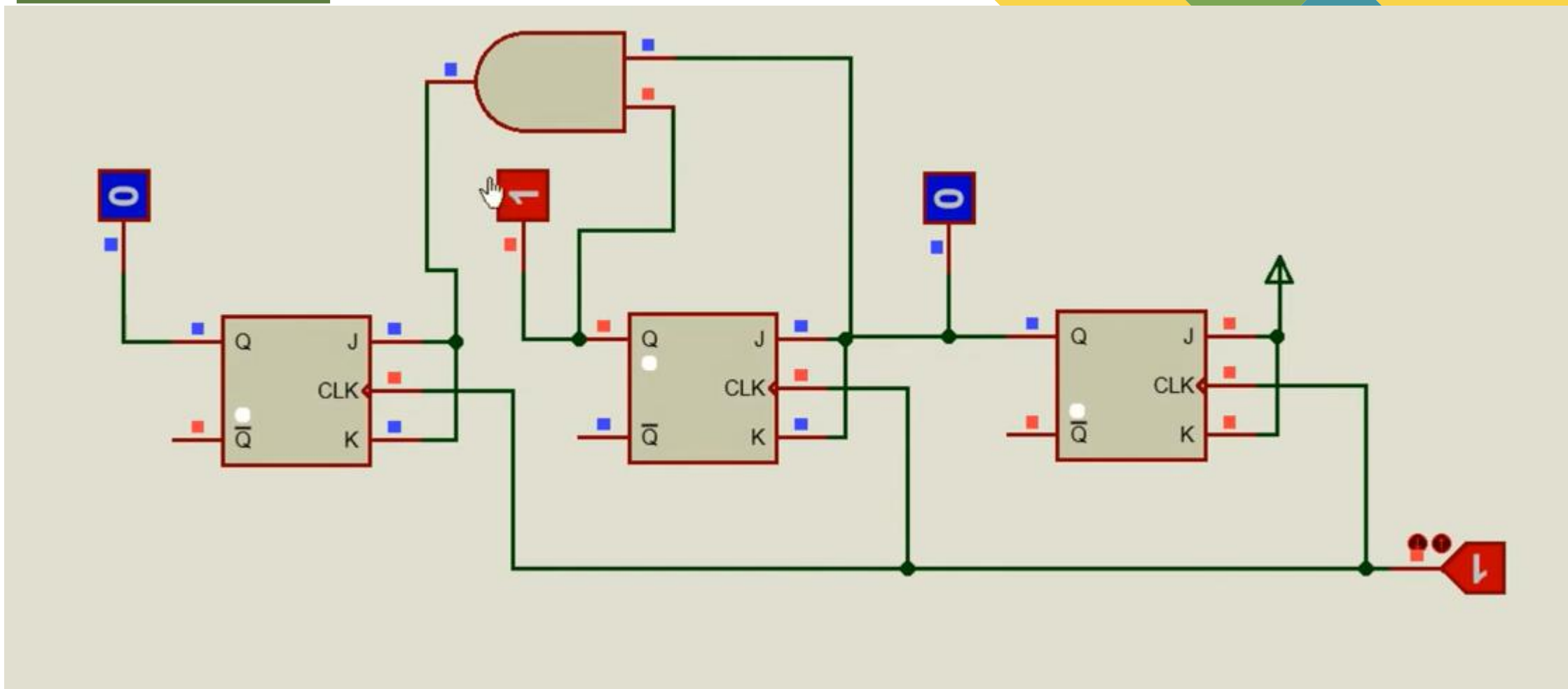
4ª Etapa

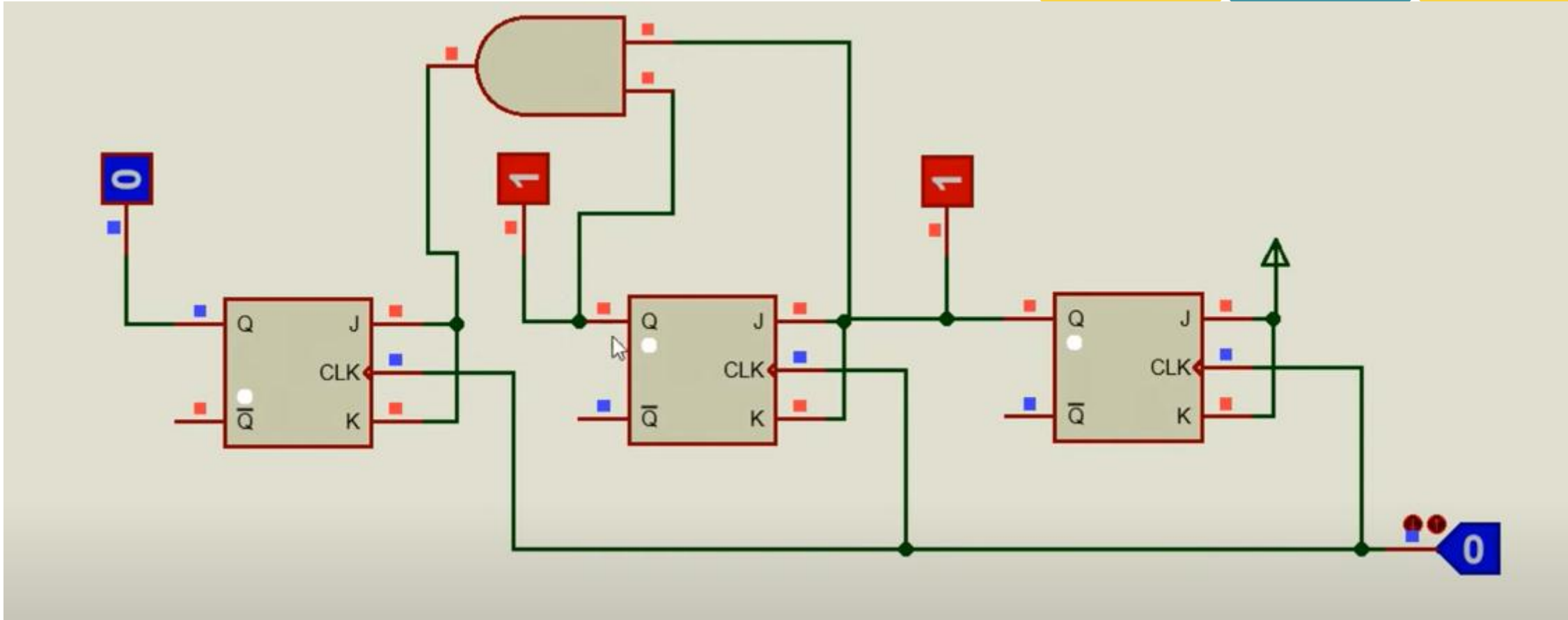


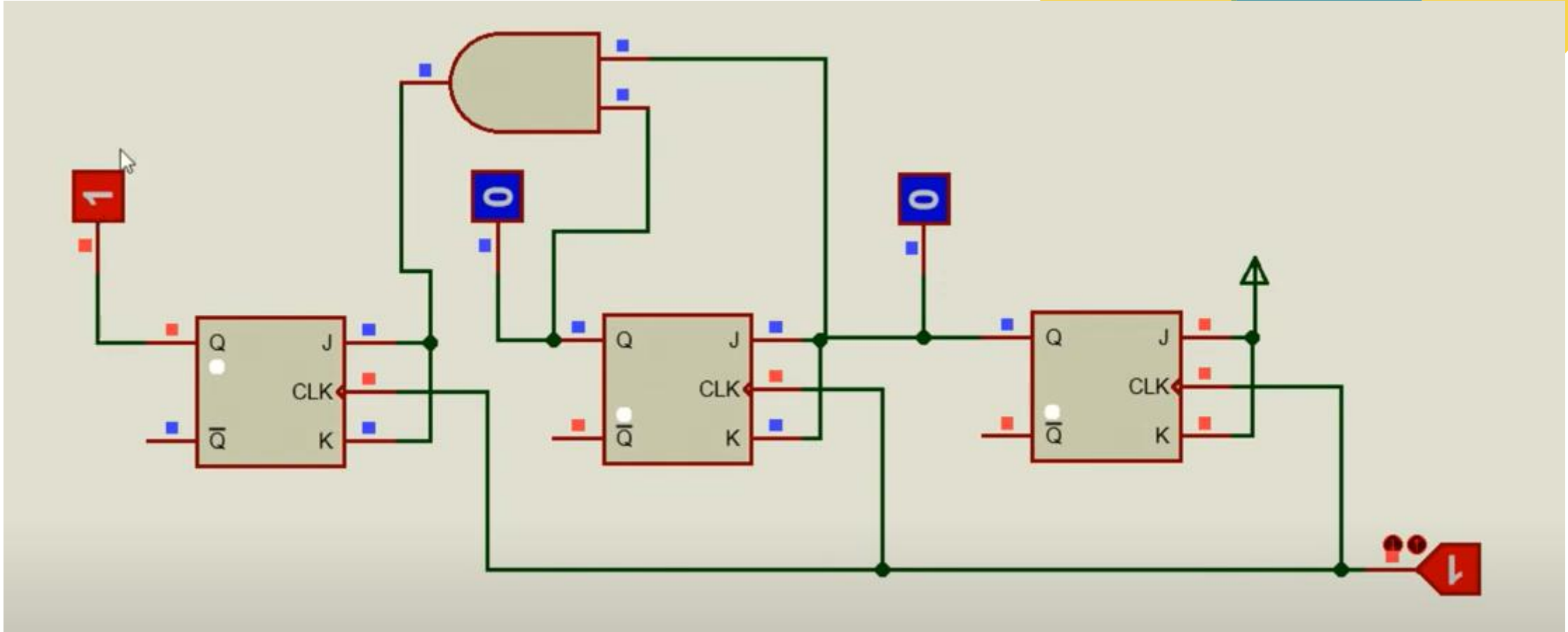




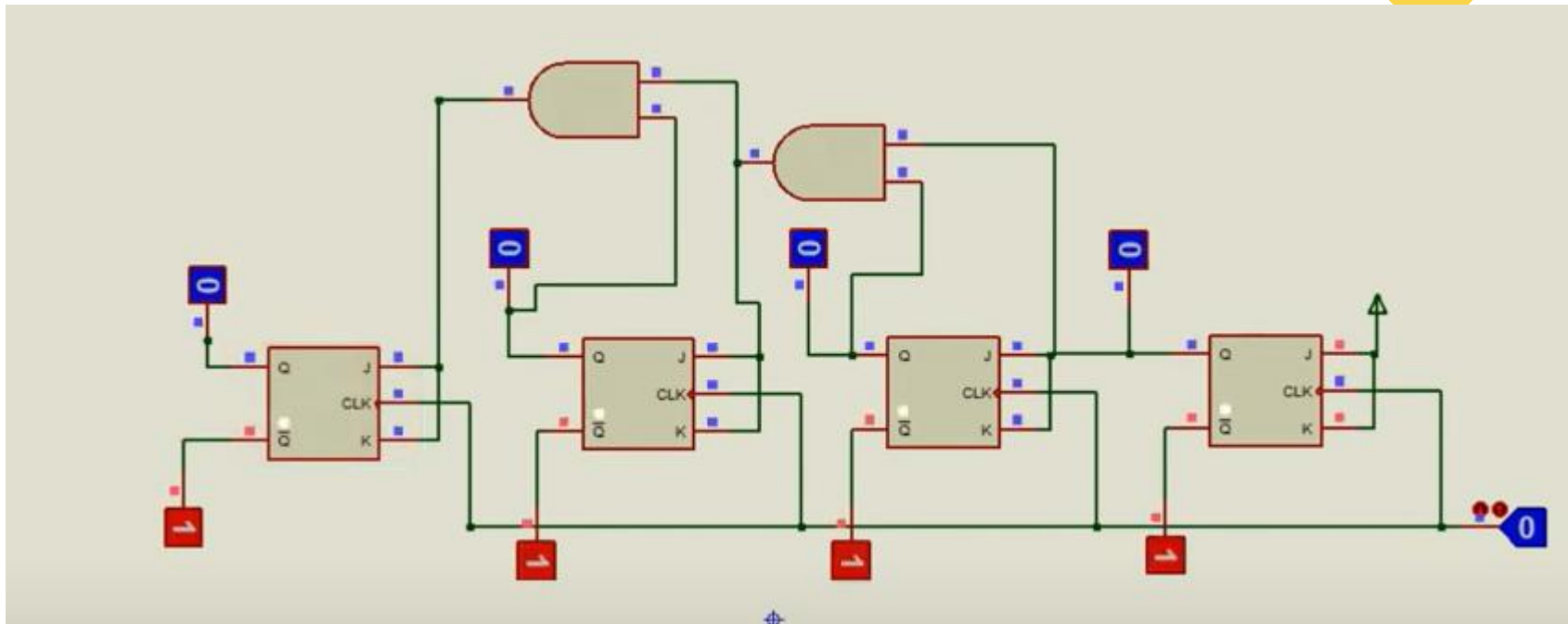


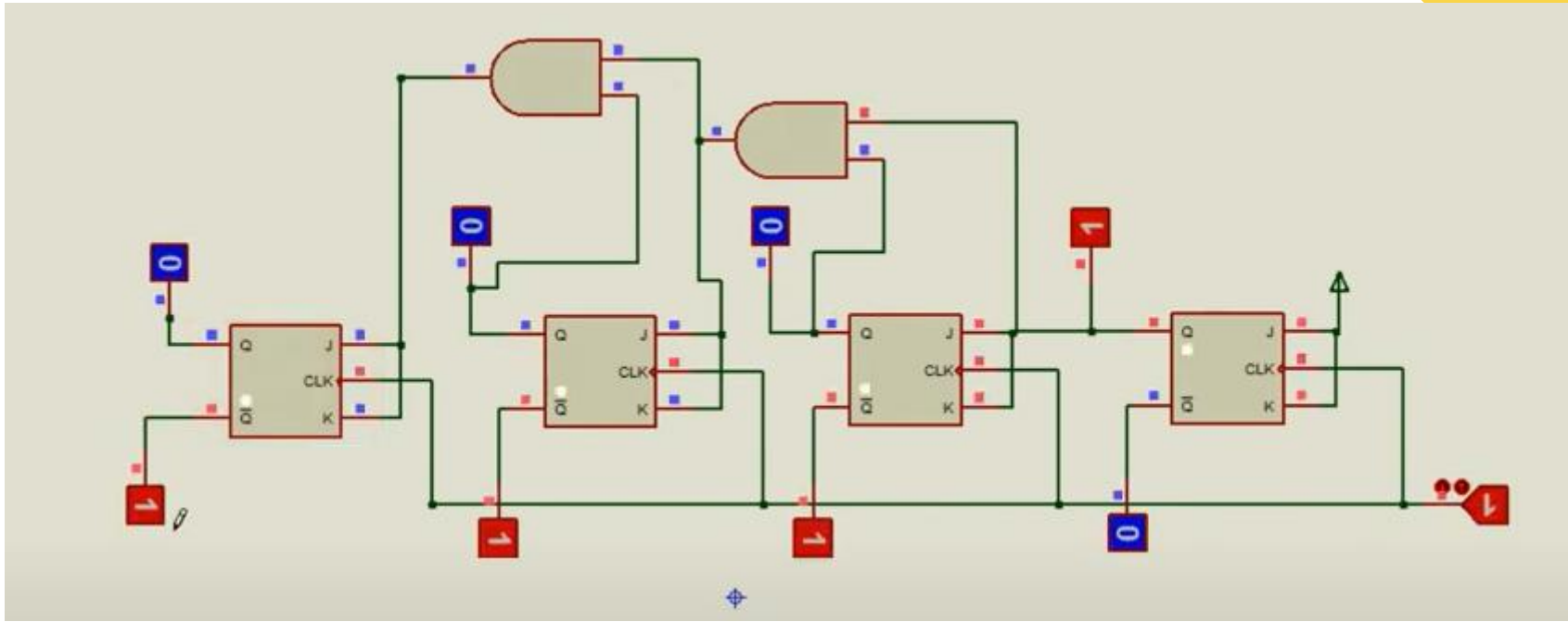


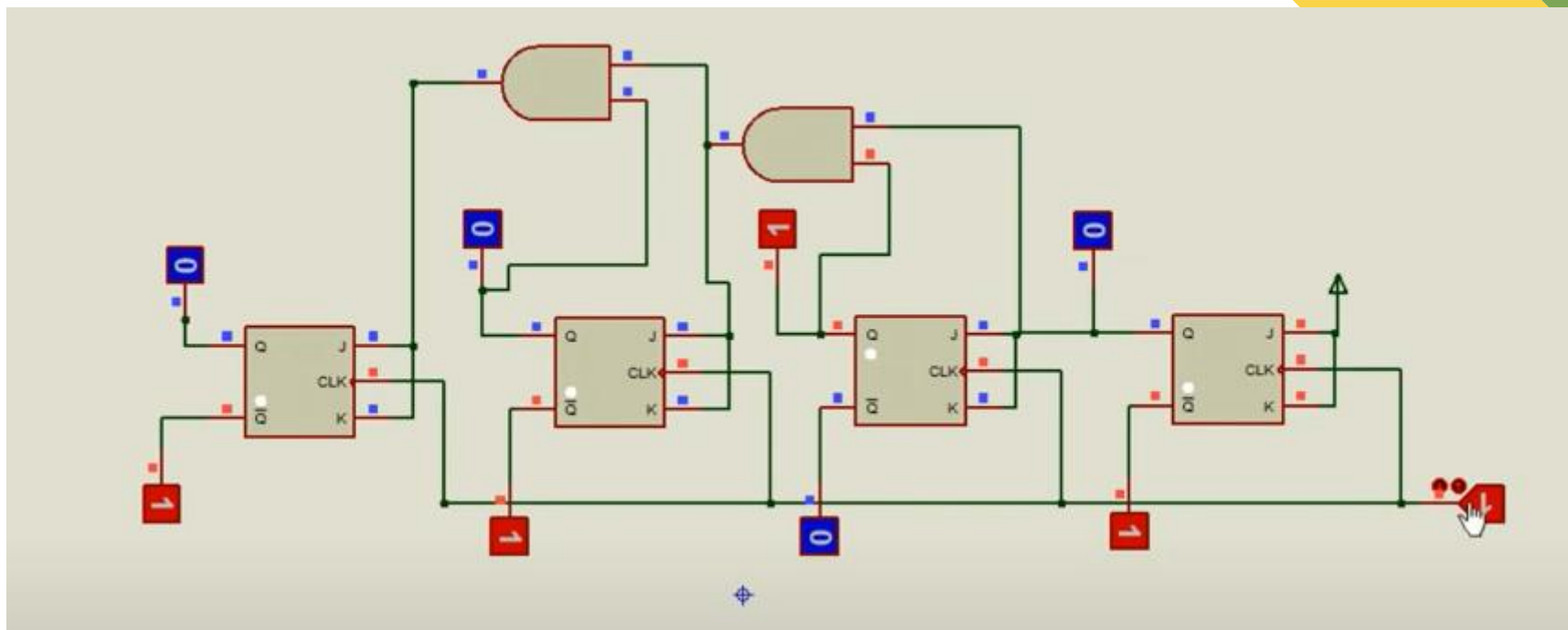




Exemplo (crescente/descrecente)







Outro esquema

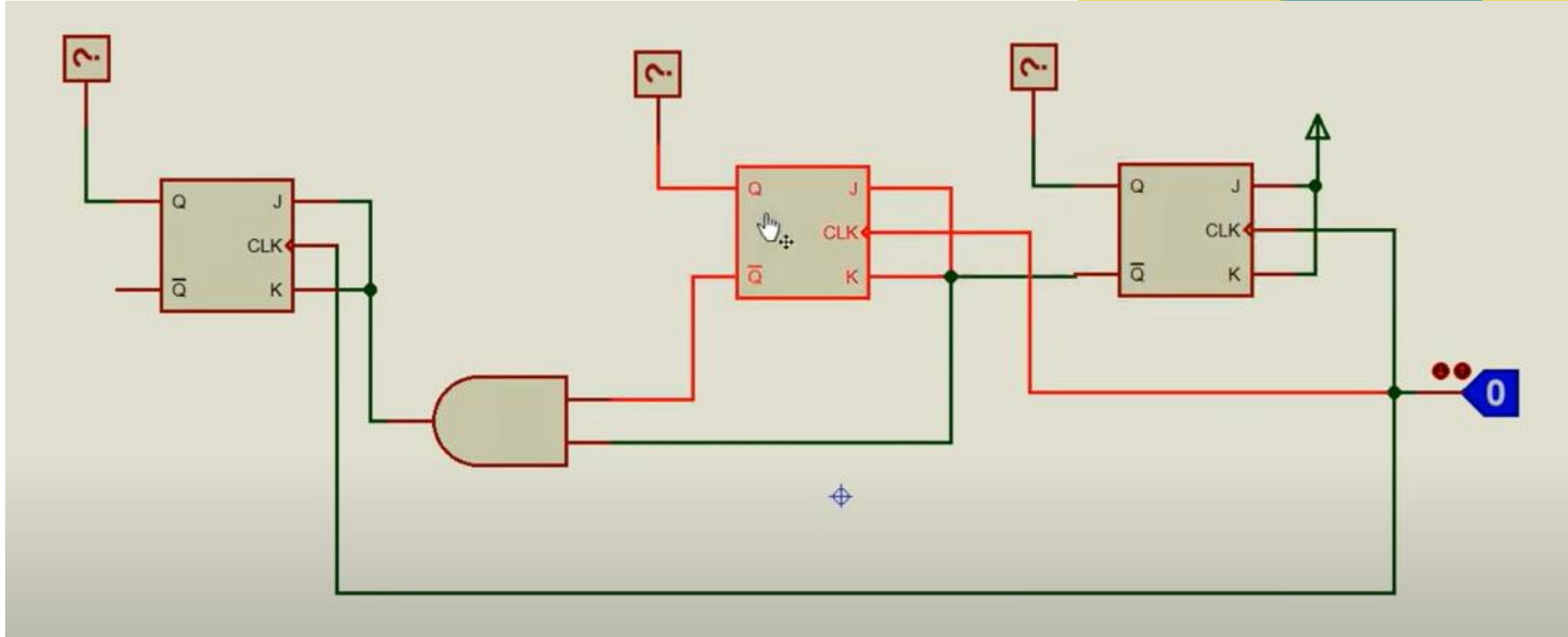




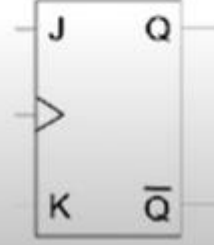
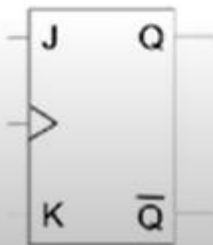
Diagrama de transição de estado

transição de Q			combinação	
			J	K
0	→	0	0	X
0	→	1	1	X
1	→	0	X	1
1	→	1	X	0

transição de Q			combinação	
			J	K
0	→	0	0	X
0	↶	1	1	X
1	↷	0	X	1
1	→	1	X	0

número	Q2	Q1	Q0	J2	K2	J1	K1	J0	K0
0	0	0	0						
3	0	1	1						
1	0	0	1						
4	1	0	0						
7	1	1	1						

$0 \rightarrow n$ $n < 2^n$



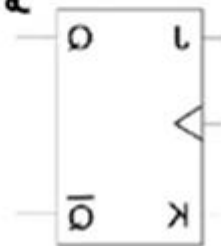
transição de Q

			combinação	
			J	K
0	→	0	0	X
0	→	1	1	X
1	→	0	X	1
1	→	1	X	0

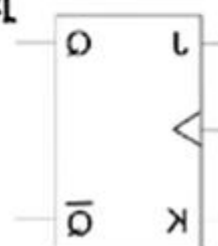
número	Q2	Q1	Q0	J2	K2	J1	K1	J0	K0
0	0	0	0						
3	0	1	1						
1	0	0	1						
4	1	0	0						
7	1	1	1						

msb

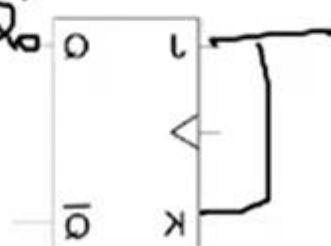
Q_2



Q_1



lsb
 Q_0



transição de Q

		combinação	
		J	K
0	→	0	X
0	→	1	X
1	→	X	1
1	→	X	0

↓

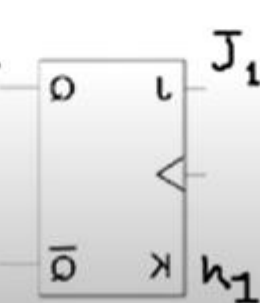
número	Q2	Q1	Q0	J2	K2	J1	K1	J0	K0
0	0	0	0	0	X				
3	0	1	1						
1	0	0	1						
4	1	0	0						
7	1	1	1						

LSB

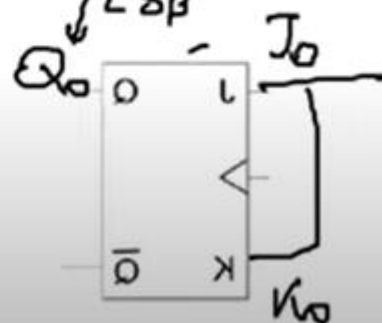
Q_2



Q_1



LSB



transição de Q

		combinação	
		J	K
0	→	0	X
0	→	1	X
1	→	X	1
1	→	X	0

número

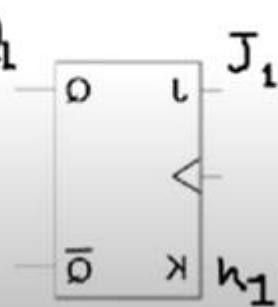
	Q2	Q1	Q0	J2	K2	J1	K1	J0	K0
0	0	0	0	0	X				
3	0	1	1	0	X				
1	0	0	1	1	X				
4	1	0	0	X	0				
7	1	1	1	X	1				

LSB

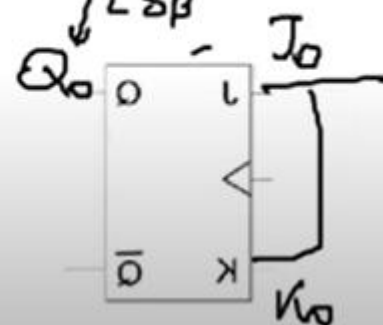
Q₂

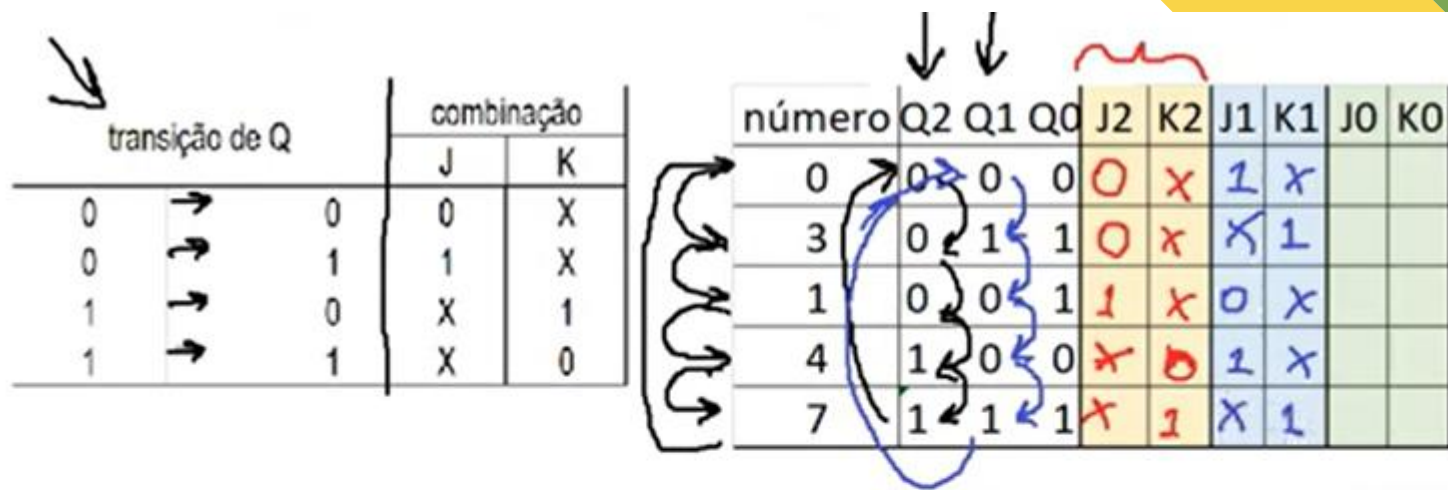


Q₁

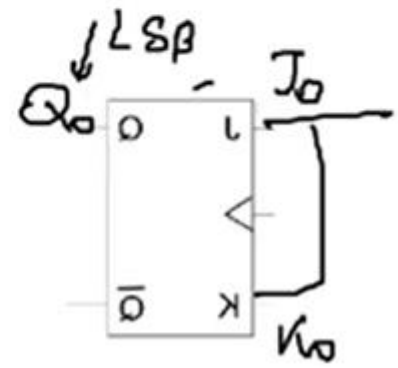
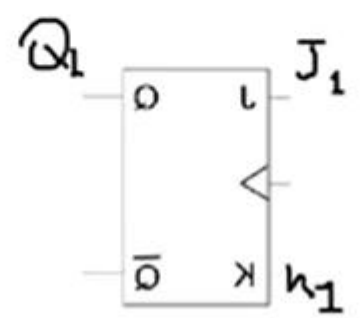
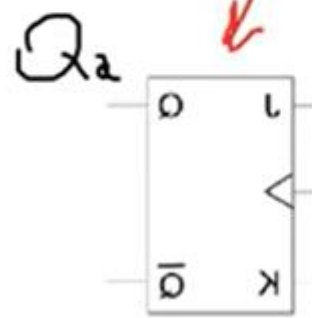


LSB





MSB



transição de Q			combinação	
			J	K
0	→	0	0	X
0	↶	1	1	X
1	↷	0	X	1
1	→	1	X	0

número	Q2	Q1	Q0	J2	K2	J1	K1	J0	K0
0	0	0	0	0	X	1	X	1	X
3	0	1	1	0	X	X	1	X	0
1	0	0	1	1	X	0	X	X	1
4	1	0	0	X	0	1	X	1	X
7	1	1	1	X	1	X	1	X	1

Aplicando mapa Karnuch

transição de Q			combinação	
			J	K
0	→	0	0	X
0	↶	1	1	X
1	↷	0	X	1
1	→	1	X	0

número	Q2	Q1	Q0	J2	K2	J1	K1	J0	K0
0	0	0	0	0	X	1	X	1	X
3	0	1	1	0	X	X	1	X	0
1	0	0	1	1	X	0	X	X	1
4	1	0	0	X	0	1	X	1	X
7	1	1	1	X	1	X	1	X	1

Q ₂ Q ₁ \ Q ₀	00	01	11	10
0				
1				

transição de Q			combinação	
			J	K
0	→	0	0	X
0	→	1	1	X
1	→	0	X	1
1	→	1	X	0

número	Q2	Q1	Q0	J2	K2	J1	K1	J0	K0
0	0	0	0	0	X	1	X	1	X
3	0	1	1	0	X	X	1	X	0
1	0	0	1	1	X	0	X	X	1
4	1	0	0	X	0	1	X	1	X
7	1	1	1	X	1	X	1	X	1

J_2

$Q_2 \backslash Q_1$	00	01	11	10
0	0			X
1	1	0	X	

$2 = X$
 $5 = X$
 $6 = X$

transição de Q			combinação	
			J	K
0	→	0	0	X
0	→	1	1	X
1	→	0	X	1
1	→	1	X	0

J_2

Q_2Q_1	00	01	11	10
0	0	X	X	X
1	1	0	X	X

número	Q2	Q1	Q0	J2	K2	J1	K1	J0	K0
0	0	0	0	0	X	1	X	1	X
3	0	1	1	0	X	X	1	X	0
1	0	0	1	1	X	0	X	X	1
4	1	0	0	X	0	1	X	1	X
7	1	1	1	X	1	X	1	X	1

$2 = X$
 $5 = X$
 $6 = X$

transição de Q			combinação	
			J	K
0	→	0	0	X
0	↶	1	1	X
1	↷	0	X	1
1	→	1	X	0

número	Q2	Q1	Q0	J2	K2	J1	K1	J0	K0
0	0	0	0	0	X	1	X	1	X
3	0	1	1	0	X	X	1	X	0
1	0	0	1	1	X	0	X	X	1
4	1	0	0	X	0	1	X	1	X
7	1	1	1	X	1	X	1	X	1
2	0	1	0						
5	1	0	1						
6	1	1	0						

transição de Q			combinação	
			J	K
0	→	0	0	X
0	↶	1	1	X
1	↷	0	X	1
1	→	1	X	0

número	Q2	Q1	Q0	J2	K2	J1	K1	J0	K0
0	0	0	0	0	X	1	X	1	X
3	0	1	1	0	X	X	1	X	0
1	0	0	1	1	X	0	X	X	1
4	1	0	0	X	0	1	X	1	X
7	1	1	1	X	1	X	1	X	1
2	0	1	0	0	X	X	1	0	X
5	1	0	1	X	1	0	X	X	1
6	1	1	0	X	1	X	1	0	X

número	Q2	Q1	Q0	J2	K2	J1	K1	J0	K0
0	0	0	0	0	X	1	X	1	X
3	0	1	1	0	X	X	1	X	0
1	0	0	1	1	X	0	X	X	1
4	1	0	0	X	0	1	X	1	X
7	1	1	1	X	1	X	1	X	1
2	0	1	0	0	X	X	1	0	X
5	1	0	1	X	1	0	X	X	1
6	1	1	0	X	1	X	1	0	X

$Q_2 Q_1$	00	01	11	10
Q_0				
0				
1				

↓

número	Q2	Q1	Q0	J2	K2	J1	K1	J0	K0
0	0	0	0	0	X	1	X	1	X
3	0	1	1	0	X	X	1	X	0
1	0	0	1	1	X	0	X	X	1
4	1	0	0	X	0	1	X	1	X
7	1	1	1	X	1	X	1	X	1
2	0	1	0	0	X	X	1	0	X
5	1	0	1	X	1	0	X	X	1
6	1	1	0	X	1	X	1	0	X

J

$Q_2 Q_1$ \ Q_0	00	01	11	10
0	0	0	X	X
1	1	0	X	X

↓

número	Q2	Q1	Q0	J2	K2	J1	K1	J0	K0
0	0	0	0	0	X	1	X	1	X
3	0	1	1	0	X	X	1	X	0
1	0	0	1	1	X	0	X	X	1
4	1	0	0	X	0	1	X	1	X
7	1	1	1	X	1	X	1	X	1
2	0	1	0	0	X	X	1	0	X
5	1	0	1	X	1	0	X	X	1
6	1	1	0	X	1	X	1	0	X

J_2

$Q_2 Q_1$	00	01	11	10
Q_0	0	0	X	X
1	1	0	X	X

$$J_2 = \overline{Q_1} \cdot Q_0$$

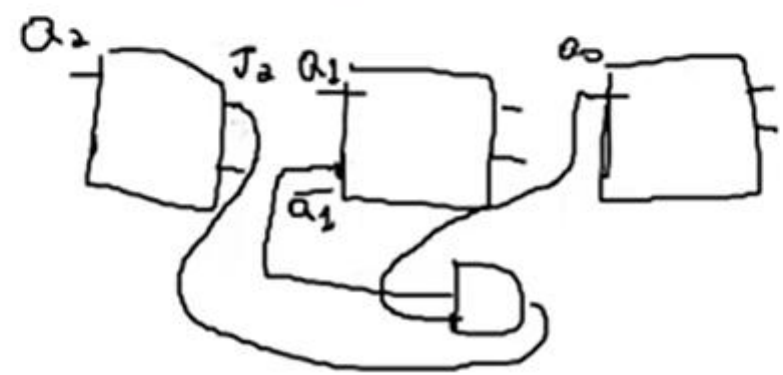
↓

número	Q2	Q1	Q0	J2	K2	J1	K1	J0	K0
0	0	0	0	0	X	1	X	1	X
3	0	1	1	0	X	X	1	X	0
1	0	0	1	1	X	0	X	X	1
4	1	0	0	X	0	1	X	1	X
7	1	1	1	X	1	X	1	X	1
2	0	1	0	0	X	X	1	0	X
5	1	0	1	X	1	0	X	X	1
6	1	1	0	X	1	X	1	0	X

J_2

Q_2Q_1 \ Q_0	00	01	11	10
0	0	0	X	X
1	1	0	X	X

$$J_2 = \overline{Q_1} \cdot Q_0$$



↓

número	Q2	Q1	Q0	J2	K2	J1	K1	J0	K0
0	0	0	0	0	X	1	X	1	X
3	0	1	1	0	X	X	1	X	0
1	0	0	1	1	X	0	X	X	1
4	1	0	0	X	0	1	X	1	X
7	1	1	1	X	1	X	1	X	1
2	0	1	0	0	X	X	1	0	X
5	1	0	1	X	1	0	X	X	1
6	1	1	0	X	1	X	1	0	X

K2

Q_2Q_1 \ Q_0	00	01	11	10
0	X	X	1	0
1	X	X	1	1

↓

número	Q2	Q1	Q0	J2	K2	J1	K1	J0	K0
0	0	0	0	0	X	1	X	1	X
3	0	1	1	0	X	X	1	X	0
1	0	0	1	1	X	0	X	X	1
4	1	0	0	X	0	1	X	1	X
7	1	1	1	X	1	X	1	X	1
2	0	1	0	0	X	X	1	0	X
5	1	0	1	X	1	0	X	X	1
6	1	1	0	X	1	X	1	0	X

K2

Q_2Q_1 \ Q_0	00	01	11	10
0	X	X	1	0
1	X	X	1	1

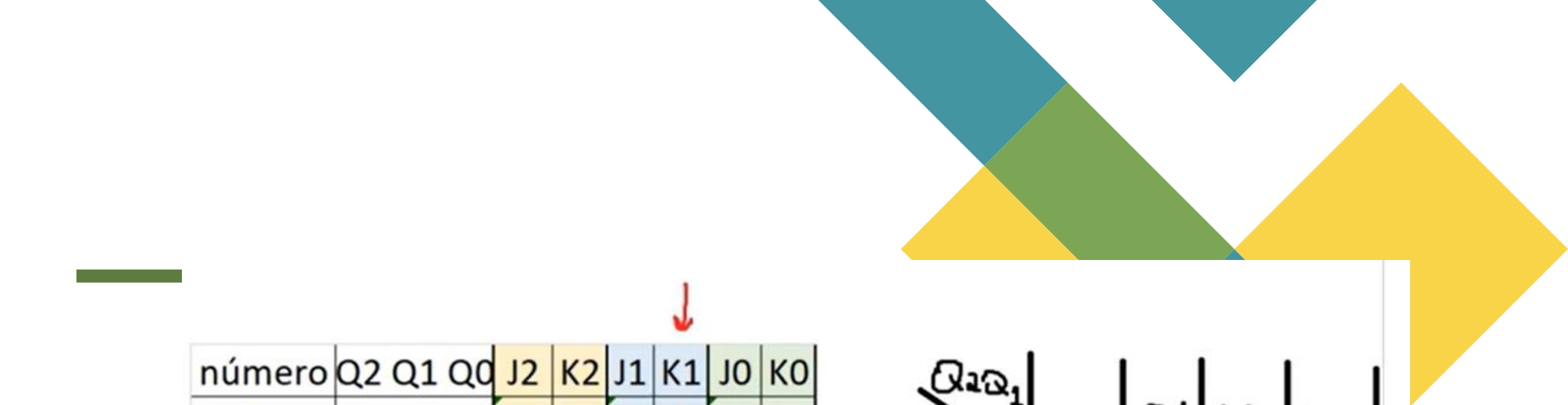
$$K_2 = Q_1 + Q_0$$

↓

número	Q2	Q1	Q0	J2	K2	J1	K1	J0	K0
0	0	0	0	0	X	1	X	1	X
3	0	1	1	0	X	X	1	X	0
1	0	0	1	1	X	0	X	X	1
4	1	0	0	X	0	1	X	1	X
7	1	1	1	X	1	X	1	X	1
2	0	1	0	0	X	X	1	0	X
5	1	0	1	X	1	0	X	X	1
6	1	1	0	X	1	X	1	0	X

Q_2Q_1	00	01	11	10
Q_0				
0	1	X	X	1
1	0	X	X	0

$$J_1 = \overline{Q_0}$$



↓

número	Q2	Q1	Q0	J2	K2	J1	K1	J0	K0
0	0	0	0	0	X	1	X	1	X
3	0	1	1	0	X	X	1	X	0
1	0	0	1	1	X	0	X	X	1
4	1	0	0	X	0	1	X	1	X
7	1	1	1	X	1	X	1	X	1
2	0	1	0	0	X	X	1	0	X
5	1	0	1	X	1	0	X	X	1
6	1	1	0	X	1	X	1	0	X

$Q_2 Q_1$	00	01	11	10
Q_0				
0	X	1	1	X
1	X	1	1	X

$K_1 = 1$



número	Q2	Q1	Q0	J2	K2	J1	K1	J0	K0
0	0	0	0	0	X	1	X	1	X
3	0	1	1	0	X	X	1	X	0
1	0	0	1	1	X	0	X	X	1
4	1	0	0	X	0	1	X	1	X
7	1	1	1	X	1	X	1	X	1
2	0	1	0	0	X	X	1	0	X
5	1	0	1	X	1	0	X	X	1
6	1	1	0	X	1	X	1	0	X

Q_1Q_0	00	01	11	10
0	1	0	0	1
1	X	X	X	1

$$J_0 = \overline{Q_1}$$

↓

número	Q2	Q1	Q0	J2	K2	J1	K1	J0	K0
0	0	0	0	0	X	1	X	1	X
3	0	1	1	0	X	X	1	X	0
1	0	0	1	1	X	0	X	X	1
4	1	0	0	X	0	1	X	1	X
7	1	1	1	X	1	X	1	X	1
2	0	1	0	0	X	X	1	0	X
5	1	0	1	X	1	0	X	X	1
6	1	1	0	X	1	X	1	0	X

K_0

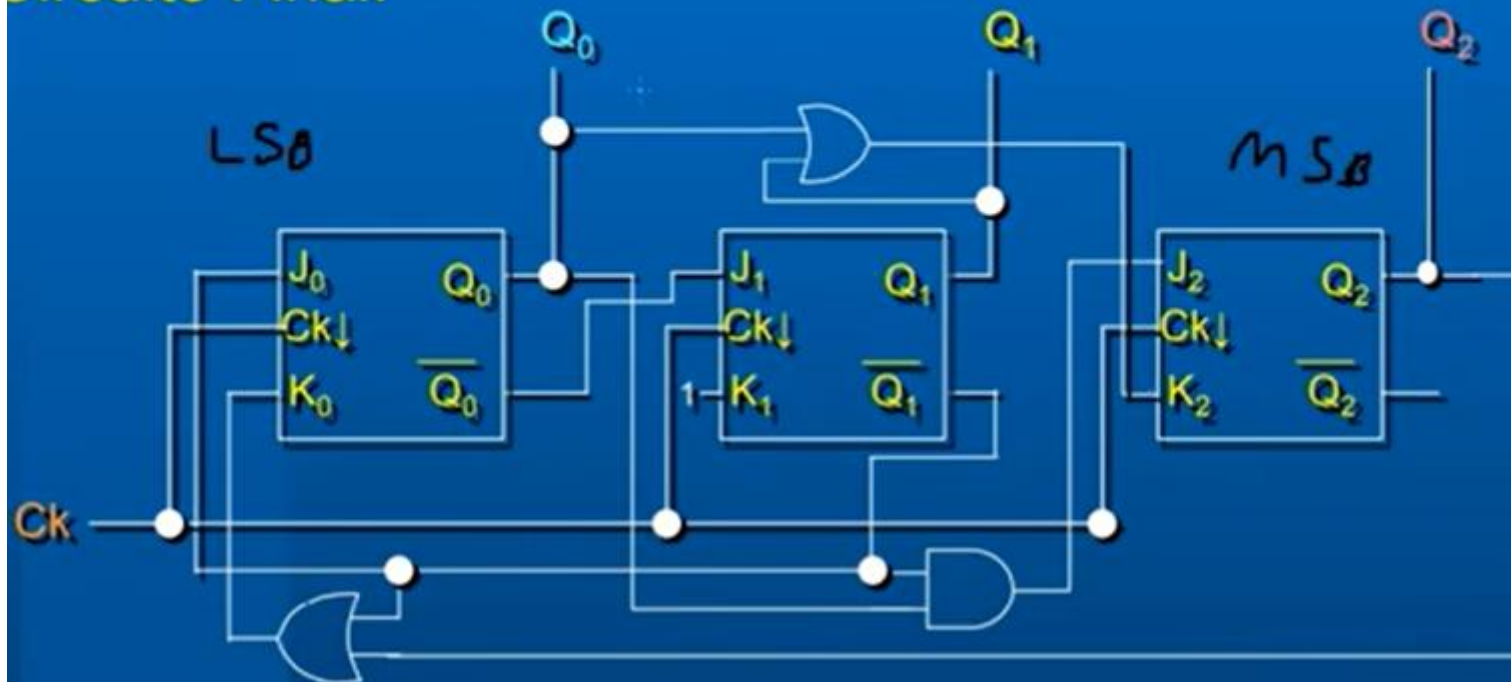
Q_2Q_1 \ Q_0	00	01	11	10
0	X	X	X	X
1	1	0	1	1

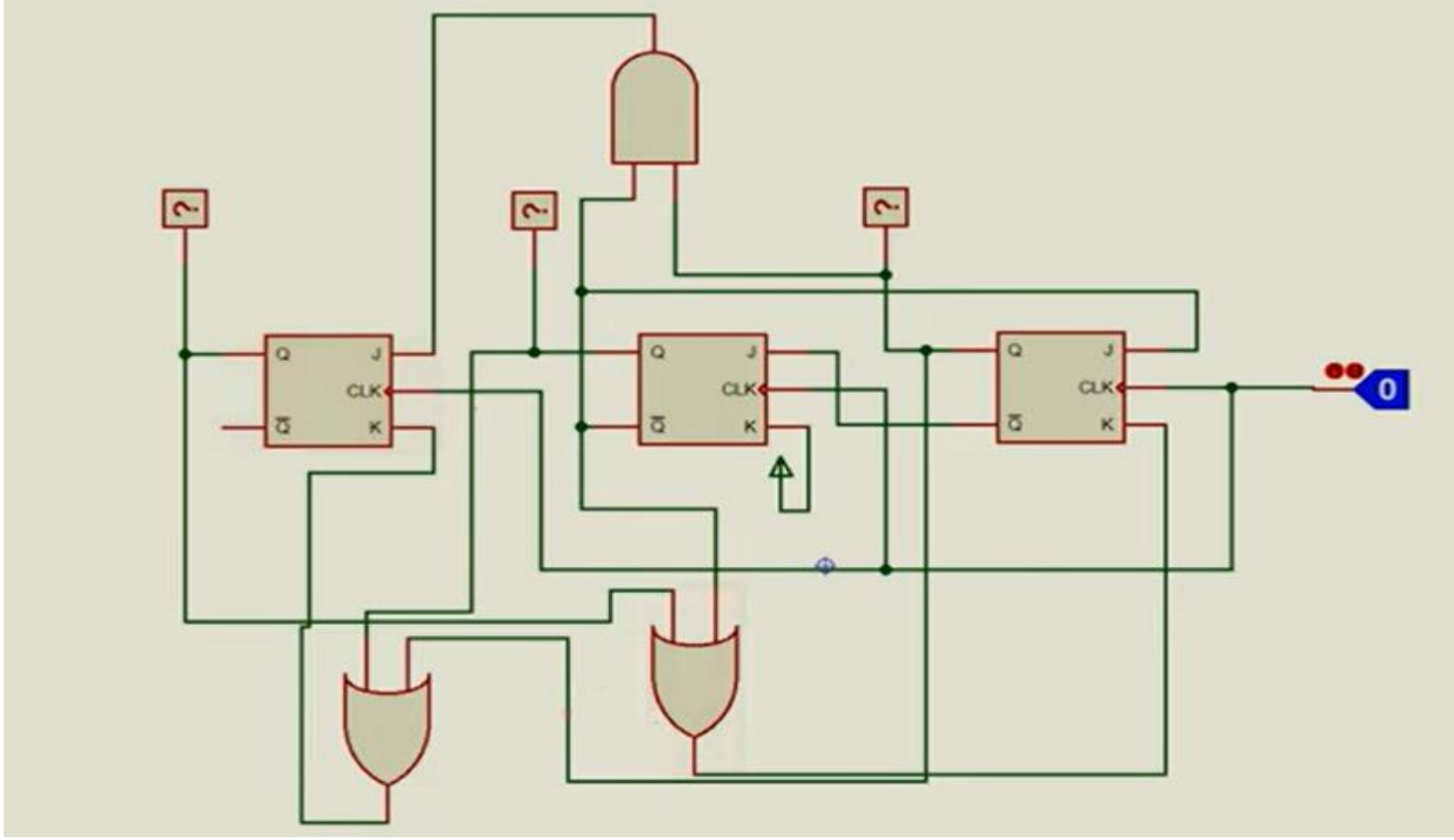
$$K_0 = Q_2 + \overline{Q_1}$$

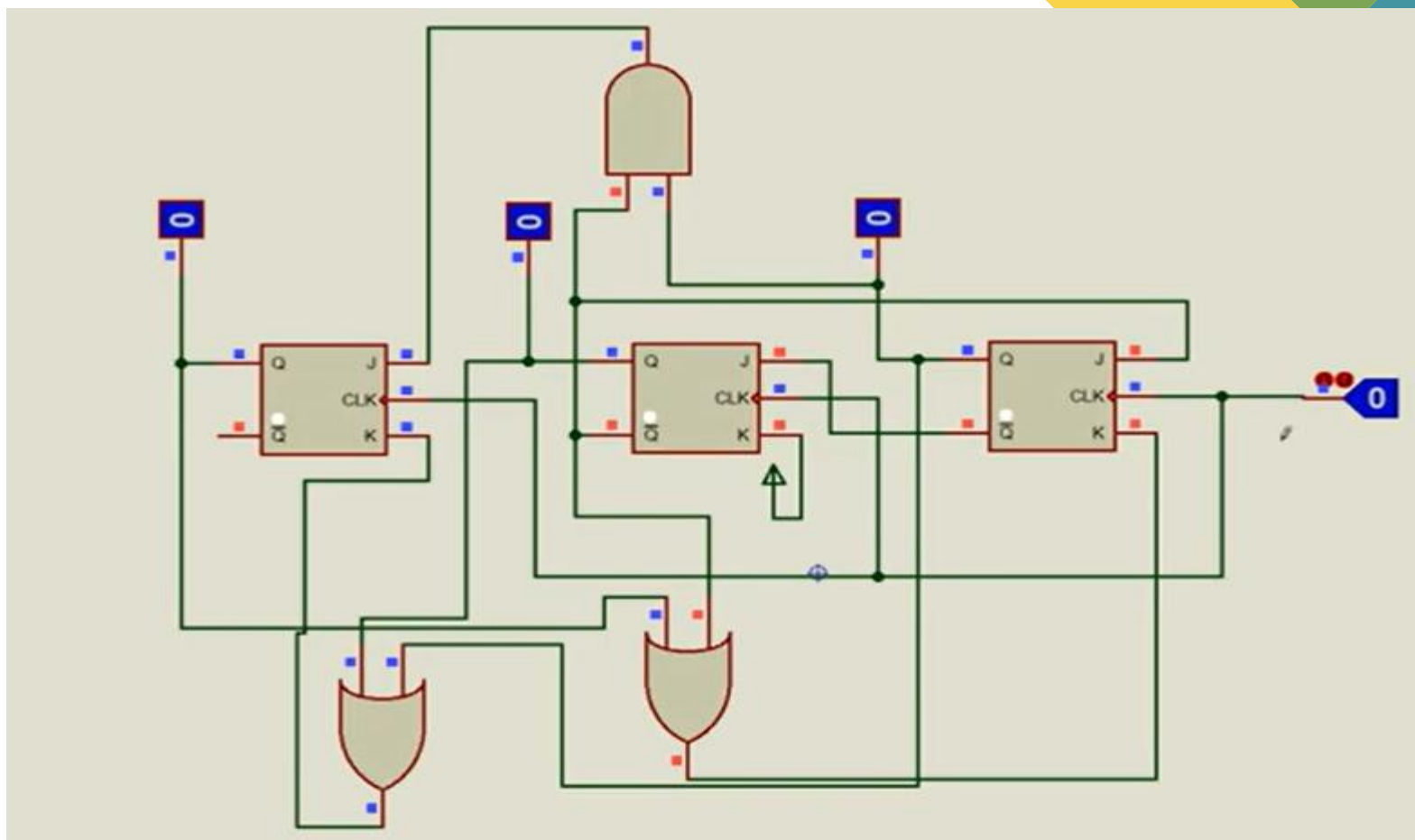
Ligações dos Flip-Flops JK:

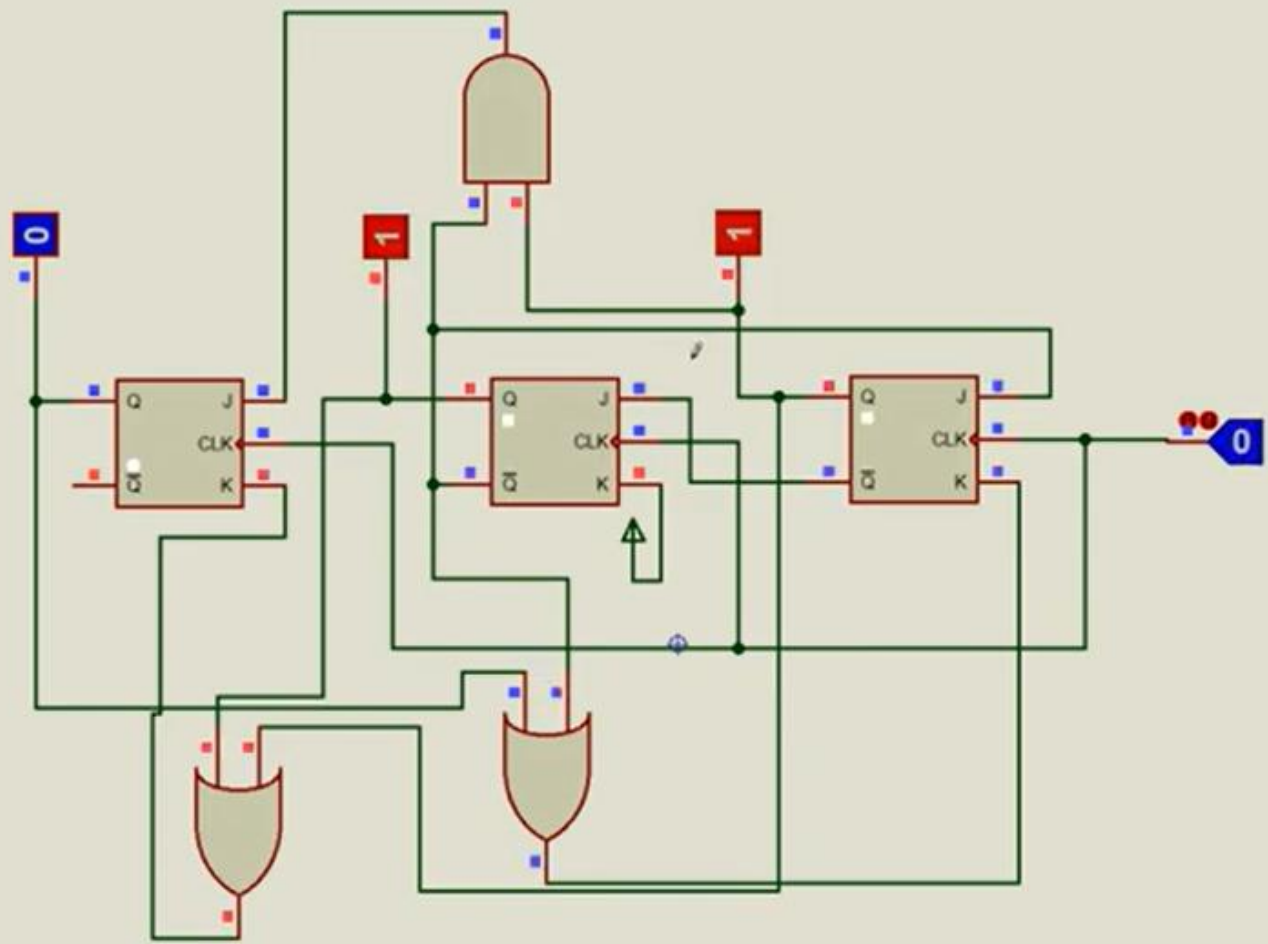
$$\begin{array}{lll} J_2 = Q_0 \overline{Q_1} & J_1 = \overline{Q_0} & J_0 = \overline{Q_1} \\ K_2 = Q_0 + Q_1 & K_1 = 1 & K_0 = Q_2 + \overline{Q_1} \end{array}$$

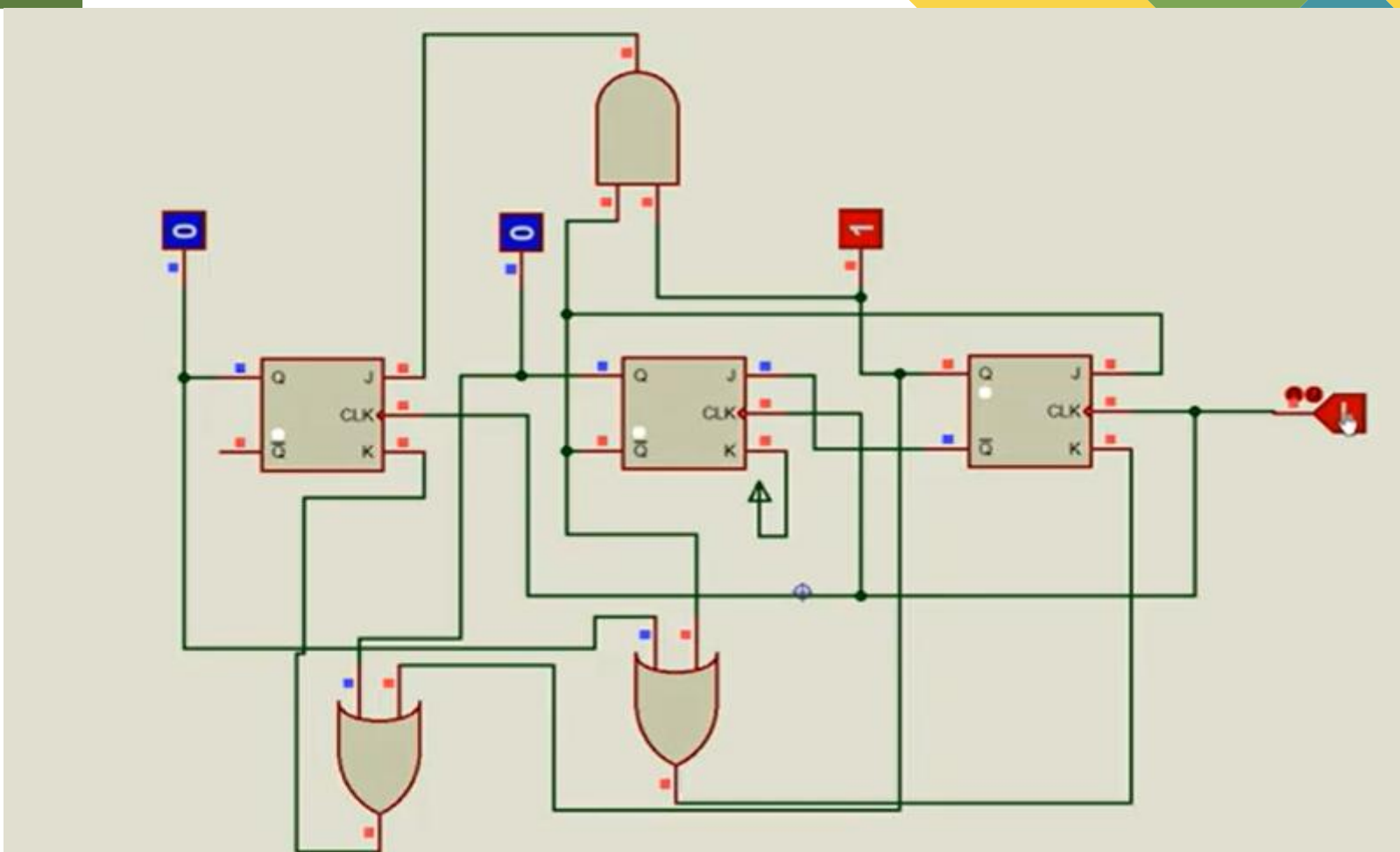
Circuito Final:

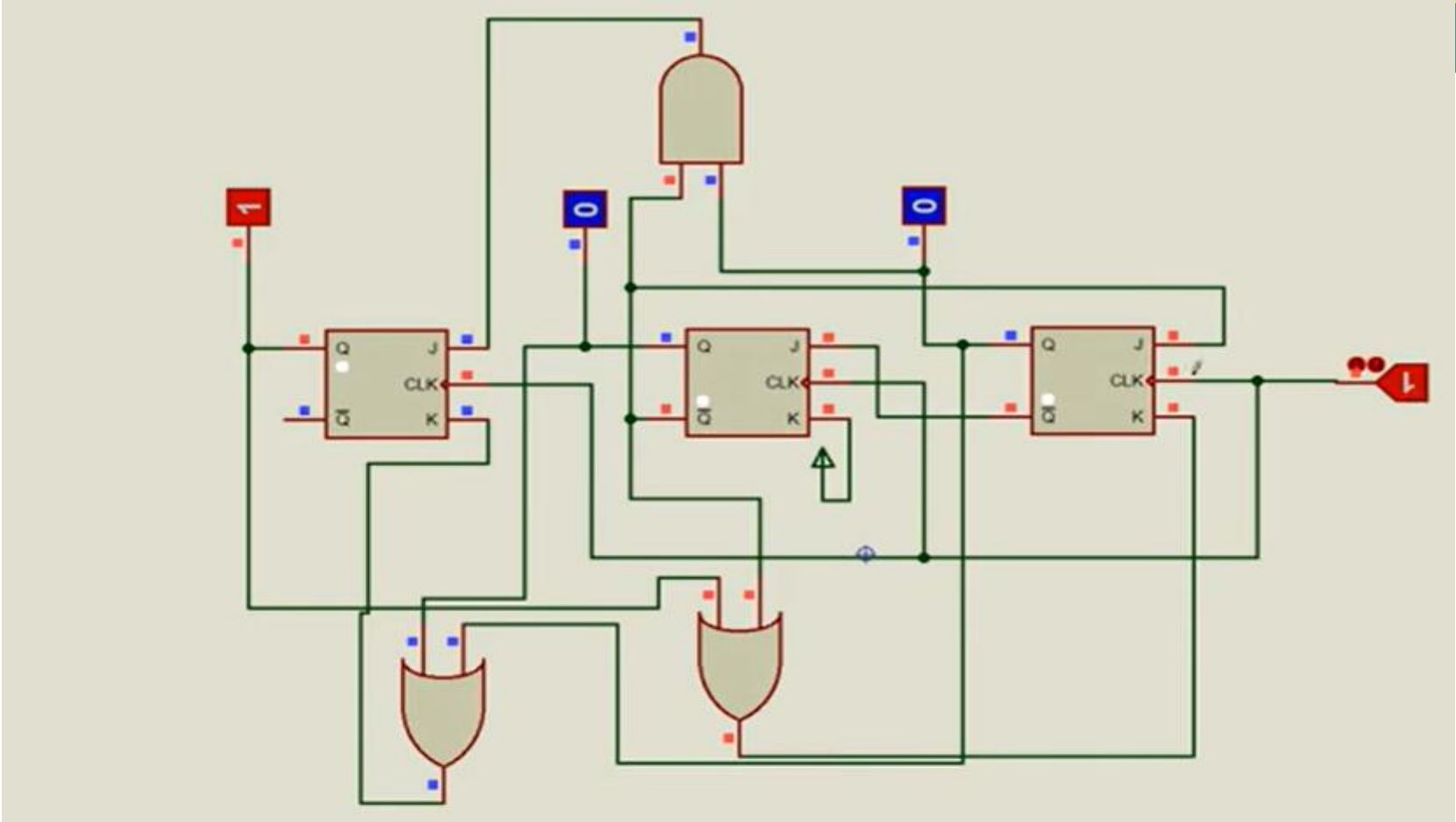


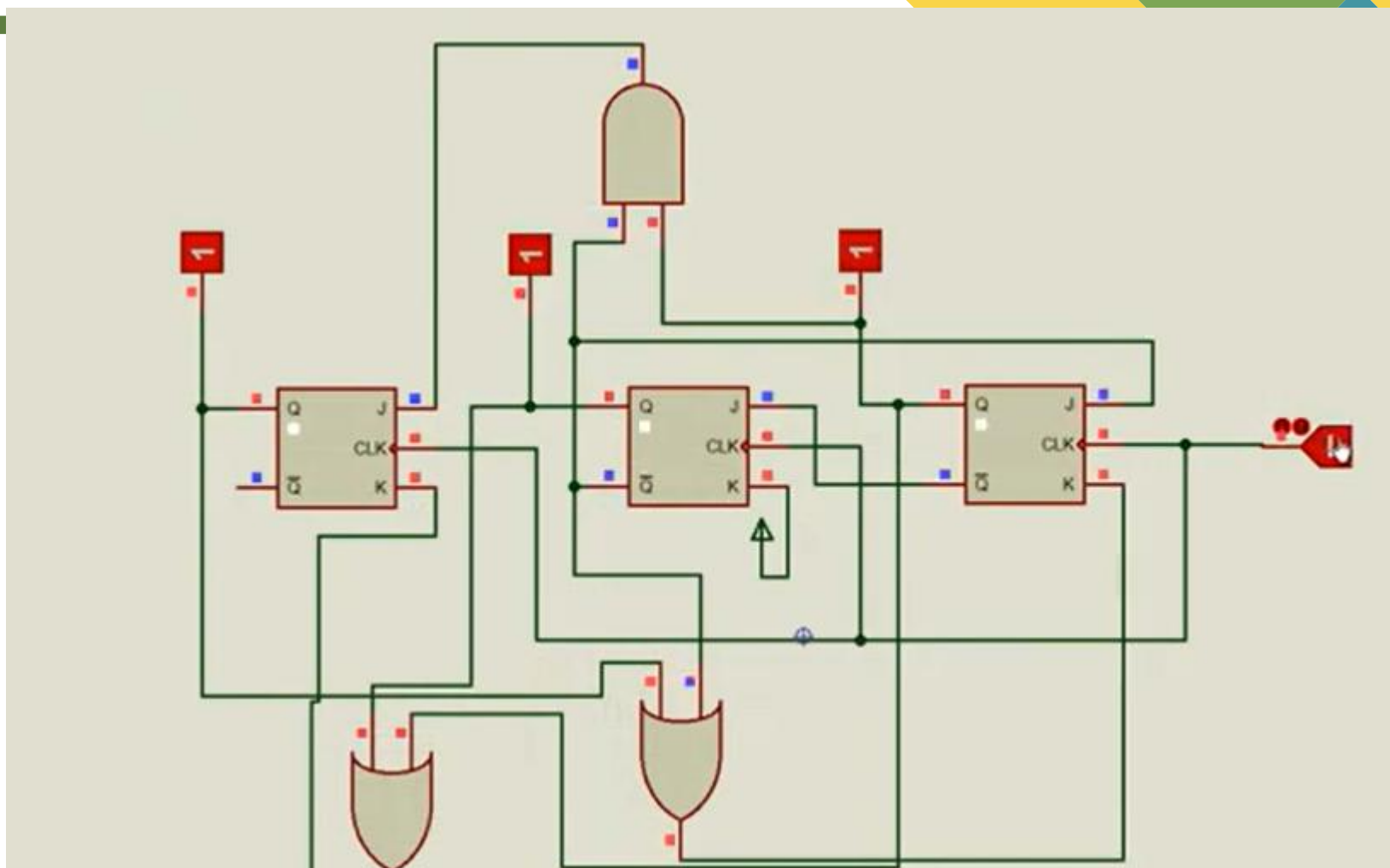


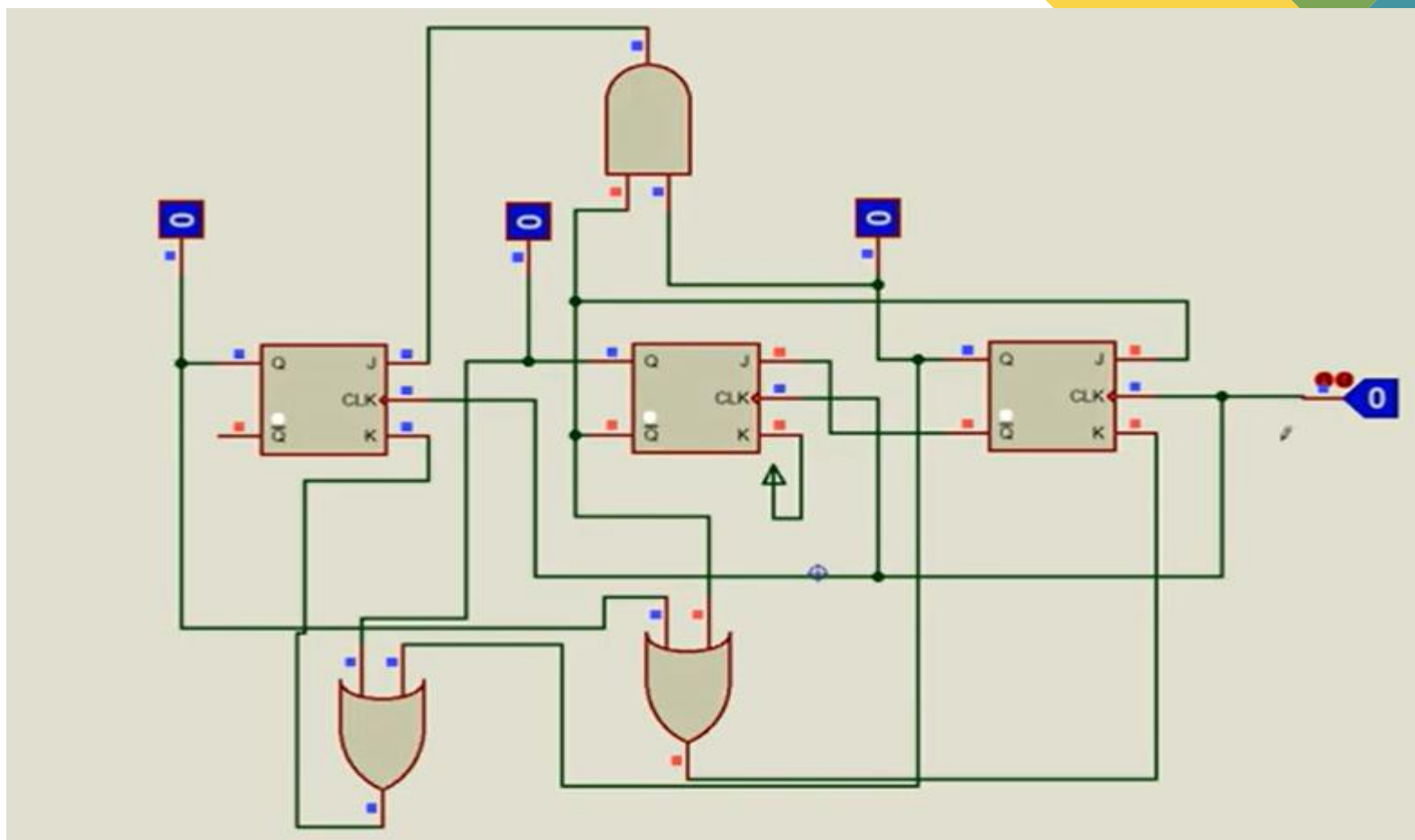












EP1 – Projetar um relógio digital (hora,min,seg)

- O aluno deve usar um circuito síncrono
- A hora, minuto e segundo deverá ser apresentado num display
- Restrições
 - Poderá usar somente portas e flip flop visto em sala de aula
- Grupo máximo de 4 alunos
- A implementação deve usar o logisim (<https://cburch.com/logisim/pt/index.html>)

Entrega

- Data 11/05
- Documentação com descritivo do circuito
 - Mapa Karnuch
 - simplificações
- Arquivo com o circuito



Atividade

- Implementar um contador de numero par

